MIL-S-62544(AT)
5 April 1988
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

SUPPRESSION, FIRE, BROMOTRIFLUOROMETHANE, HALON 1301 SYSTEMS

This specification is approved for use within the US Army Tank-Automotive Command , Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Halon 1301 fire suppression systems, referred to herein as "system(s)", used in military vehicles to automatically detect and quench fires. The system's major components consist of optical fire sensor assemblies (OFSA'S) electronically coupled to the fixed fire extinguishers through a standard electronic control module (MSEC). The fixed fire extinguishers, referred to herein as "extinguishers" and "valve and cylinder assemblies", also have manual release capabilities (see 6.1).

1.2 Classification. The systems shall be of the following types and classes, as specified (see 6.2):

Type I - Discriminatory. OFSA(S) will not respond to high intensity, short duration emissions caused by chemical energy munition penetration of aluminum and steel armor. The quantity of OFSA'S required shall be as specified (see 6.2 and 6.7.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of-use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - Non-discriminatory. OFSA(s) will respond to chemical energy armor penetrations. The quantity of OFSA'S required shall be as specified (see 6.2 and 6.7.2).

Class 1 - Valve and cylinder assembly(s) containing 2.2 kilograms (kg) [5 pounds (lb)] of Halon 1301 in a 2360 cubic centimeter (cm^3) [144 cubic inch (in^3)] receiver cylinder. 1/

Class 2 - Valve and cylinder assembly(s) containing 3.18 kg (7 lb) of Halon 1301 in a 3340 cm^3 (204 in^3) receiver cylinder. 1/

Class 3 - Valve and cylinder assembly(s) containing 4.54 kg (10 lb) of Halon 1301 in a 4720 cm^3 (288 in^3) receiver cylinder. 1/

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-E-489	- Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).
W-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-H-46170	Hydraulic Fluid, Rust Inhibited, Fire Resistant Synthetic Hydrocarbon Base.
MIL-M-62545	Module , Standard Electronic Control.
MIL-S-62546	Sensor, Fire, Optical.
MIL-V-62547	Valve and Cylinder Assemblies, Halon.

1/ Quantity of each size to make a system shall be as specified (see 6.2).

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STANDARDS
MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-171	Finishing of Metal and Wood Surfaces.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1275	Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	Calibration Systems Requirements.
MS3114	Connectors, Receptacle, Electric, Solder Type, Rear Mounting Jam Nut, Bayonet Coupling.
MS3121	Connectors, plug, Electric, Crimp Type, Cable Connecting,Bayonet Coupling.
MS3449	Connectors, Receptacle, Electric, Series 2) Single Hole Mount, Bayonet Coupling, Solder Pin Contact, Class H.
MS3474	Connectors, Receptacle, Electrical, Series 2, Crimp Type, Narrow Flange Mount, Bayonet Coupling, Classes A, L, S and W.
MS28889	Valve, Air, High-Pressure Charging.

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

DRAWINGS
ARMY

11669713	- Safety Disc.
12288692	- Pin, Safety.
12314282	- Optical Fire Sensor Assembly.
12314288	- Amplifier, Standard Control Electronic.
12314494	- Valve and Bottle Assembly.
12314510	- Pressure Gauge, 0-1500 PSIG, Temperature Compensated/Coordinated.
12343131	- Antirecoil Assembly.
12343165	- Pin, Safety.
12343265	- Connector, Electrical Receptacle.

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PUBLICATIONS
TEST PROCEDURES

Nuclear Survivability Criteria	- Nuclear Survivability Criteria for Automatic Optical Fire Detection and Suppression Systems for Combat Vehicles (c).
TACOM Technical Report No. 12679	- EMI Test for Automatic Fire Suppression System, Environmental Test Procedures.

[Copies of specifications, standards, drawings and publications required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity. (see 6.9)]

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The system furnished under this specification shall be a system which is authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Materials used shall be as specified herein and as specified in referenced specifications, standards and drawings. Materials shall be suitable over an ambient temperature range of minus (-) 51 to plus (+) 71 degrees Celsius (“C) [-60 to +160 degrees Fahrenheit (“F)]; -51 to +125°C (-60 to +257°F) for materials in the OFSA. There shall be no changes in physical or dimensional characteristics that could result in a system malfunction. Materials shall be free of defects which adversely affect performance or serviceability of the system (see 4.8.1).

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which could encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing approved protective finishes, coatings or insulation between the mating materials and shall be in accordance with provisions of MIL-STD-171 which address reduction of corrosion at intermetallic contact points. Corrosion-resistant materials shall be used whenever possible (see 4.8.1).

3.2.2 Encapsulation. Encapsulation, if used, shall be of silicone rubber compounds to permit removal for repair (see 4.8.1).

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3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.7.4).

3.3 Design and construction. Unless otherwise specified (see 6.2), the system shall be a two shot, two extinguisher per shot system (see 6.7.7). A complete system comprises of OFSA(s) electronically linked to extinguisher(s) through appropriate harnesses and connectors, a MSEC and an alarm panel. Interconnecting cables and the alarm panel shall be as specified by the contracting officer (see 6.2). The fire suppression systems specified herein do not cover the interconnecting cables or the alarm panel. These components are peculiar to individual vehicle installations, and are described herein only for reference. Figures 1, 2, and 3 provide block diagrams, for reference, illustrating cabling variations for systems with the maximum numbers of OFSA'S and extinguishers. Extinguishers shall be capable of discharge by both electrical and mechanical means. The system shall be designed to operate over an ambient temperature range of -51 to + 71°C (-60 to +160°F), except that the OFSA shall be capable of operating between -51 and +125°C (-60 and +257"F). The suppression system shall be composed of components , sub-assemblies and parts called out on applicable drawings, and shall include major components depicted on the drawings and standards specified in table I. The physical size, input and output orientations and mounting provisions for the assemblies shall comply with the envelope requirements shown on Drawings 12314288 (MSEC), 1231482 (OFSA) and 12314494 (valve and cylinder assembly). Component assemblies shall require no adjustment or calibration as a condition for use in different vehicle applications (see 4.8.1 and 4.8.2).

TABLE I, System components.

Drawing number	Standard number	Component
12314288		MSEC .
12314282		OFSA.
12314494		Valve and cylinder assembly.
12343131		Antirecoil device.
12314510		Pressure gauge.
11669713		Safety disc.
12343165		Safety pin. 1/
	MS28889	Nitrogen fill valve.
	MS3474	Standard electrical connectors.

1/ Optional with 12288692

3.3.1 Components. Systems shall be comprised of the following major components : The OFSA, the MSEC and the valve and cylinder assembly. These component assemblies may be acquired from the same or different manufacturers whose names appear on QPL's, and are intended to operate as "black box" components (see 4.8.1).

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3.3.1.1 Qualified products. Any component of this system, when listed on a QPL shall be furnished from a source listed or approved for listing on the QPL in accordance with the applicable specification or drawing at the time set opening of bids. Documentation of the item acquisition from the qualified source shall be made available to the Government for review. The documentation shall include the QPL date, identification of the supplier, purchase order and quantity (see 4.1.3 and 4.8.1).

3.3.1.2 OFSA.

3.3.1.2.1 Type I system. OFSA'S in type I systems shall meet the requirements specified in MIL-S-62546 for type I OFSA'S (see 4.8.1).

3.3.1.2.2 Type II system. OFSA'S in type II systems shall meet the requirements specified in MIL-S-62546 for type II OFSA'S (see 4.8.1).

3.3.1.3 MSEC. MSEC'S shall meet the requirements of MIL-M-62545 (see 4.8.1).

3.3.1.4 Extinguishers.

3.3.1.4.1 Class 1 system. Valve and cylinder assemblies in class 1 systems shall meet the requirements specified in MIL-V-62547 for class 1 extinguishers (see 4.8.1).

3.3. 1.4.2 Class 2 system. Valve and cylinder assemblies in class 2 systems shall meet the requirements specified in MIL-V-62547 for class 2 extinguishers (see 4.8.1).

3.3. 1.4.3 Class 3 system. Valve and cylinder assemblies in class 3 systems shall meet the requirements specified in MIL-V-62547 for class 3 extinguishers (see 4.8.1).

3.3.1.4.4 Dip or syphon tubes. Dip or syphon tubes shall not be used except when specified by the contracting officer (see 6.2).

3.3.2 Connectors. Connectors on components shall be as specified on Drawings 12314282, 12314288 and as specified below (see 4.8.1 and 4.8.2):

a. OFSA connector: MS3474 W12-8PN, MS3449 H12-8PN, MS3114 W12-8PN or 12343265-1.
b. MSEC connector J1: MS3474 W22-21PN, MS3121 W22-21PN, MS3114 W22-21PN or 12343265-2.
co MSEC connector J2: MS3474 W22-21SN, MS3121 W22-21SN, MS3114 W22-21SN or 12343265-3.
d. MSEC connector J3: MS3474 W18-32SN, MS3121 W18-32SN, MS3114 W18-32SN or 12343265-4.

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3.3.3 Electrical interface. Connector pins shall be component electrical interface points for in and out signals as defined in table II and as specified in tables III through VI (see 4.8.3).

3.3.4 Interchangeability. All components of the system shall be completely and functionally interchangeable without modification or adjustment ,'see 4.8.3).

TABLE II. Electrical signal characteristics. 1/ 2/

A. LOGIC SIGNALS
10 to 30 V dc Source output impedance < 2000 ohms 3/ Load input impedance ≥ 20 000 ohms 4/ Quiescent condition: 1 V dc maximum Input impedance (source and load) > 20 000 ohms
B. EXTINGUISHER DRIVE SIGNALS
16 to 30 V dc 10 A minimum for load impedance ≤ 1.3 ohms 15 A maximum Total load impedance: 0.5 to 2.3 ohms Quiescent condition: see signal C
C. CONTINUITY MONITORING SIGNALS
5 mA maximum Source input and output impedance > 20 000 ohms Total load impedance: ≤ 2.3 ohms
D. LAMP DRIVE SIGNALS
16 to 30 V dc Source input impedance > 20 000 ohms Load impedance ≥ 750 ohms Quiescent condition: 1 V dc maximum Source input and output impedance > 20 000 ohms

1/ All outputs shall be capable of withstanding short circuits of indefinite duration.

2/ Whenever a range of values is specified for an input signal or input duration, the component shall be capable of meeting all requirements at any point within the specified range.

3/ Less than or equal to (≤).

4/ Greater than or equal to (≥).

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TABLE III. OFSA connector J3. 1/

Pin	Function	In/Out	Signal (see table II)	Duration
A	POWER.	In	15 to 30 V dc in accordance with MIL-STD-1275	
B	RETURN.	In		
C	LARGE FIRE SIGNAL. Produced when (1) Sensing a fire which has reached the sensitivity threshold (size of fire as defined in 3.4.1.2), or (2) Following receipt of BITE signal (Pin E).	Out	Signal A	(1) As long as fire exceeds threshold, or (2) For not less than 5 ms and not more than 4.0 s following receipt of BITE signal (Pin E)
D	SMALL FIRE SIGNAL. Produced when (1) Sensing a fire which has reached the sensitivity threshold (size of fire as defined in 3.4.1.1), or (2) Following receipt of BITE signal (Pin E).	Out	Signal A	(1) As long as fire exceeds threshold, or (2) For not less than 5 ms and not more than 4.0 s following receipt of BITE signal (Pin E)
E	BITE SIGNAL. A signal received at this pin shall initiate a test of OFSA functions and shall result in output signals from Pins C and D.	In	Signal A	5 ms to 4.0 s
F and G	CODE PINS. These pins are connected together within the wiring harness for vehicles with aluminum armor. With steel armor, the pins are not used.			
H	UNUSED			

1/ OFSA connector (to MSEC), Connector MS3474 W12-8PN.

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TABLE IV. OFSA connections (J3). 1/

Pin	Function	In/Out	Signal (see table II)	Duration
A	POWER TO OFSA NUMBER 1.	Out	15 to 30 V dc 150 mA minimum 15 mA minimum (quiescent)	
B	RETURN FROM OFSA NUMBER 1.	Out		
C	LARGE FIRE SIGNAL FROM OFSA NUMBER 1.	In	Signal A	See table III, Pin C
D	SMALL FIRE SIGNAL FROM OFSA NUMBER 1.	In	Signal A	See table III, Pin D
E	BITE SIGNAL TO OFSA NUMBER 1.	Out	Signal A	5 ms to 4.0 s
J	POWER TO OFSA NUMBER 2.	Out	15 to 30 V dc 150 mA minimum 15 mA minimum (quiescent)	
K	RETURN FROM OFSA NUMBER 2.	Out		
L	LARGE FIRE SIGNAL FROM OFSA NUMBER 2.	In	Signal A	See table III, Pin C
M	SMALL FIRE SIGNAL FROM OFSA NUMBER 2.	In	Signal A	See table III, Pin D
N	BITE SIGNAL TO OFSA NUMBER 2.	Out	Signal A	5 ms to 4.0 s
T	POWER TO OFSA NUMBER 3.	Out	15 to 30 V dc 150 mA minimum 15 mA minimum (quiescent)	
U	RETURN FROM OFSA NUMBER 3.	Out		
V	LARGE FIRE SIGNAL FROM OFSA NUMBER 3.	In	Signal A	See table III, Pin C
W	SMALL FIRE SIGNAL FROM OFSA NUMBER 3.	In	Signal A	See table III, Pin D
X	BITE SIGNAL TO OFSA NUMBER 3.	Out	Signal A	5 ms to 4.0 s

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TABLE IV. OFSA connections (J3). 1/ - Continued.

Pin	Function	In/Out	Signal (see table II)	Duration
b	POWER TO OFSA NUMBER 4.	Out	15 to 30 V dc 150 mA minimum 15 mA minimum (quiescent)	
c	RETURN FROM OFSA NUMBER 4.	Out		
d	LARGE FIRE SIGNAL FROM OFSA NUMBER 4.	In	Signal A	See table III, Pin C
e	SMALL FIRE SIGNAL FROM OFSA NUMBER 4.	In	Signal A	See table III, Pin D
f	BITE TO OFSA NUMBER 4.	Out	Signal A	5 ms to 4.0 s
F	UNUSED			
G	UNUSED			
H	UNUSED			
P	UNUSED			
R	UNUSED			
S	UNUSED			
Y	UNUSED			
Z	UNUSED			
a	UNUSED			
g	UNUSED			
h	UNUSED			
j	UNUSED			

1/ MSEC connector J3 (to OFSA), Connector MS3474 W18-32SN.

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TABLE V. Extinguisher connection (J2). 1/

Pin	Function	In/Out	Signal (See table II)	Duration
A	EXTINGUISHER NUMBER 1 ACTIVATION SIGNAL AND CONTINUOUS CONTINUITY MONITOR.	Out	Signal B and C	33 to 38 ms
B	EXTINGUISHER NUMBER 1 RETURN.			
C	EXTINGUISHER NUMBER 1 PRESSURE/FLOW MONITOR.	Out	High pressure and no flow-short to Pin B Low pressure or flow-open circuit to Pin B	
F	EXTINGUISHER NUMBER 2 ACTIVATION SIGNAL AND CONTINUOUS CONTINUITY MONITOR.	Out	Signal B and C	33 to 38 ms
G	EXTINGUISHER NUMBER 2 RETURN.			
H	EXTINGUISHER NUMBER 2 PRESSURE/FLOW MONITOR.	Out	High pressure and no flow-short to Pin G Low pressure or flow-open circuit to Pin G	
L	EXTINGUISHER NUMBER 3 ACTIVATION SIGNAL AND CONTINUOUS CONTINUITY MONITOR.	Out	Signal B and C	33 to 38 ms
M	EXTINGUISHER NUMBER 3 RETURN.			
N	EXTINGUISHER NUMBER 3 PRESSURE/FLOW MONITOR.	Out	High pressure and no flow-short to Pin M Low pressure or flow-open circuit to pin M	
S	EXTINGUISHER NUMBER 4 ACTIVATION SIGNAL AND CONTINUOUS CONTINUITY MONITOR.	Out	Signal B and C	33 to 38 ms
T	EXTINGUISHER NUMBER 4 RETURN.			
U	EXTINGUISHER NUMBER 4 PRESSURE/FLOW MONITOR.	Out	High pressure and no flow-short to Pin T Low pressure or flow-open circuit to Pin T	
D	UNUSED			
E	UNUSED			
J	UNUSED			
K	UNUSED			
P	UNUSED			
R	UNUSED			
V	UNUSED			
W	UNUSED			
X	UNUSED			

1/ MSEC connector J2 (to extinguishers), Connector MS3474 W22-21SN.

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TABLE VI. Test and alarm panel connection (J1). 1/

Pin	Function	In/Out	Signal (See table II)	Duration
A	POWER.	In	16 to 30 V dc in accordance with MIL-STD-1275	
N	RETURN.	In	40 A maximum	
B	AUXILIARY POWER. May be used as power source for test and alarm panel or auxiliary devices.	Out	16 to 30 V dc 40 A maximum	
M	RETURN.	Out		
C	EXTINGUISHER NUMBER 1 STATUS LAMP SIGNAL.	Out	Signal D	Continuous when continuity is lost at Pins A or C, table V
D	EXTINGUISHER NUMBER 2 STATUS LAMP SIGNAL.	Out	Signal D	Continuous when continuity is lost at Pins F or H, table V
E	EXTINGUISHER NUMBER 3 STATUS LAMP SIGNAL.	Out	Signal D	Continuous when continuity is lost at Pins L or N, table V
F	EXTINGUISHER NUMBER 4 STATUS LAMP SIGNAL.	Out	Signal D	Continuous when continuity is lost at Pins S or U, table V
G	MANUAL RELEASE SIGNAL FROM TEST AND ALARM PANEL. Causes release of first shot. Second signal received after inhibit period of 4 to 6 s causes release of second shot.	In	10 to 30 V dc	
H	BITE ACTIVATION SIGNAL FROM TEST AND ALARM PANEL. Initiates a test of sensing system.	In	10 to 30 V dc	

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TABLE VI. Test and alarm panel connection (J1). 1/ - Continued.

Pin	Function	In/Out	Signal (See table II)	Duration
J	BITE "PASS" SIGNAL. Produced after complete system test has been accomplished and system is completely functional.	Out	Signal D	4 to 6 s
K	BITE "FAIL" SIGNAL TO TEST AND ALARM PANEL. Produced if "BITE" finds a fault.	Out	Signal D	Until retest results in a pass indication
L	FIRE WARNING LAMP SIGNAL. Produced whenever an appropriate signal is received at any of the following pins in connector J3: D, M, W or e.	Out	Signal D	As long as appropriate signal is received at Pin D, M, W, or e of connector J3
T	VENTILATION FAN ACTIVIATION SIGNAL. Produced 8 to 10 s after extinguisher activation signal.	Out	16 to 30 V dc (suitable for operating a relay) source input impedence $\geq$ 20 000 ohms (Signal D)	4 to 6 s
U	AUDIBLE FIRE WARNING SIGNAL. Produced whenever an appropriate signal is received at any of the following pins in connector J3: D, M, W, e or C, V, L, d.	Out	16 to 30 V dc (suitable for operating a relay) source input impedence $\geq$ 20 000 ohms (Signal D)	As long as appropriate signal is received at Pin D, M, W, e, or C, L, V, d of connector J3
P	OFSA 1 FAULT SIGNAL.	Out	Signal D	Same as Pin K
R	OFSA 2 FAULT SIGNAL.	Out	Signal D	Same as Pin K
S	UNUSED			
V	OFSA 3 FAULT SIGNAL.	Out	Signal D	Same as Pin K
W	OFSA 4 FAULT SIGNAL.	Out	Signal D	Same as Pin K
X	MSEC FAULT SIGNAL.	Out	Signal D	Same as Pin K

1/ MSEC connector J1 (to test and alarm panel), Connector MS3474 W22-21PN.

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3.3.5 System built-in-test-equipment (BITE). The system shall have a BITE capability which includes a check (go/no-go) of the OFSA'S and MSEC (initiated by press and release of the appropriate test and alarm panel button). Generation of a fault signal shall not render the remainder of the system inoperative. The complete system check shall be accomplished in accordance with the following sequence (see table VI) (see 4.8.3):

a. Test signal received by MSEC from test and alarm panel.
b. MSEC inhibits extinguisher activation signals for duration of test.
c. MSEC sends test signal to each OFSA.
d. OFSA'S generate optical stimuli (at least one of which shall provide check of dirty sensor windows) for each sensing element causing each OFSA to generate both small fire and large fire output signals.
e. If the MSEC receives correct fire signals from all OFSA'S and MSEC circuitry is functioning properly, then a "pass" signal shall be provided to the appropriate output pin. If the correct signals are not received within 5 seconds (s) from all OFSA'S or if a fault exists within the MSEC circuitry, "fail" signals (one per faulty component and one for the system) shall be provided to the appropriate output pin(s) until a retest indicates that the faults has been corrected. The complete system test shall be completed and the system shall be fully functional within 10 s.

3.3.6 Polarity. The system shall operate with a negatively grounded two wire electrical circuit (see 4.8.3).

3.4 Performance. The system shall be capable of detecting hydrocarbon mist or surface fires, differentiating between large and small fires and providing an appropriate output response. Response to a small fire shall be as specified in 3.4.1.1. Response to a large fire shall be the generation of extinguisher drive signals for appropriately connected fire extinguishers in addition to appropriate small fire signals. Type I systems shall not respond to high intensity, short duration, short duration emissions caused by chemical energy munition penetration of aluminum and steel armor. Specific response requirements are given in subsequent sections. Shown for reference in figure 4 is a characteristics optical emission spectrum for an open hydrocarbon fire within an envelope of theoretical graybody radiation curves. Figure 5 is also a characteristic optical emission spectra for an open hydrocarbon fire. The spectra should be considered typical of optical emissions for which the system shall respond.

3.4.1 Response characteristics. The fire sensing system shall respond to radiation stimuli as indicated in the following paragraphs (see 4.8.4.1).

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3.4.1.1 Early fire warning (small fire). The fire sensing system shall respond as specified herein in not more than 5 s to a 130-millimeter (mm) [5-inch (in)] diameter pan-fire, 76 mm (3 in) minimum depth containing 840 cm^3 (51 in^3) of diesel fuel (DF-2 of W-F-800) at a distance of l2OO mm (48 in). The MSEC early fire warning circuit shall energize the fire warning lamp and audible fire warning signal pins when a fire reaches the small fire threshold, without producing an extinguisher drive signal. There shall be one fire warning circuit per MSEC. The fire warning circuit shall not latch (see table VI) (see 4.8.4.1.1).

3.4.1.2 Secondary sensitivity threshold (large fire). The fire sensing system shall respond as specified herein in not more than 5 s to the hydrocarbon pan-fire described in 3.4.1.1, except the distance shall be 380 mm (15 in) from the centerline of the pan to the OFSA. The MSEC shall respond as required when a large fire signal is supplied for 1 to 2 s to appropriate pins within MSEC connector J3 (see table IV) (see 4.8.4.1.2).

3.4.1.3 Response time. The valve shall begin to discharge Halon within 10 milliseconds (ins) after receiving an activation signal. The extinguishers shall completely discharge the liquid phase of Halon within the time limits specified below for each class of extinguishers (see MIL-V–62547), while at an ambient temperature of +25 plus or minus (+) 8°C (+77 + 14°F) (see 4.8.4.1.3):

a. Class 1, maximum discharge time 135 ms.
b. Class 2, maximum discharge time 190 ms.
c. Class 3, maximum discharge time 270 ms.

3.4.1.4 Penetration response time (type I system only). The system, consisting of the OFSA(s), MSEC and alarm panel, shall respond within 6 ms to a hydrocarbon fire ignited subsequent to and resulting from penetration of steel and aluminum armor by chemical energy munitions. Conditions shall be in accordance with 4.8.4. 1.4 with the OFSA located not less than 380 mm (15 in) from the fire (see 4.8.4.1.4).

3.4.2 Start-up. The system shall not false alarm upon the application of 18 and 30 volt (V) step inputs to the power pins (initial power-on condition) (see 4.8.4.2 and 6.7.3).

3.4.3 Input and output power. System power requirements include the power utilized by the system itself in the quiescent and operating modes and the power that would be used by one or more test and alarm panels or similar devices. The vehicle battery and vehicle master switch power will be made available. Block diagrams of a total system are shown in figures 1, 2 and 3 (see 4.8.4.3).

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3.4.3.1 Input power. The system shall meet the response requirements over an input voltage range (to the MSEC) of 16 to 30 V direct current (de). (Normal operation will be at 18 to 30 V dc. Voltage may fall to 16 V dc during cranking.) (see 4.8.4.3.1).

3.4.3. 1.1 Quiescent condition. In the quiescent mode of operation (standby surveillance), no more than 140 milliamperes (mA) nor less than 15 mA current draw at 24 V dc nominal shall be required for the MSEC. In addition to the 140 mA, current draw shall not exceed 15 mA nor be less than 7.5 mA at 24 V dc nominal per OFSA. Other input power (up to 320 mA at 24 V dc nominal) may be required for energizing system lamps, if in the circuit. The discharge valve solenoid circuit and the pressure/flow circuit shall be capable of accepting continuity monitoring signals of 5 mA continuously without valve activation. The resistance of the solenoid and the pressure/flow circuits shall not exceed 8 ohms (see 4.8.4.3.1.1).

3.4.3.1.2 Input voltage decay. The system shall not false alarm (produce small or large fire output signals from either the MSEC or any OFSA) when the system is undergoing the input voltage decays resulting from prolonged operation without recharging of the battery (see 4.8.4.3.1.2).

3.4.3. 1.3 Steady state voltage extremes. The system shall sustain no damage from the application of any steady state voltage up to 40 V dc (see 4.8.4.3.1.3).

3.4.3.1.4 BITE power. During a complete BITE system test, power consumption shall at no time exceed 15 watts (W). If extinguisher drive signals are checked during BITE [using internal extinguisher simulator loads which draw 10 to 15 amperes (A)] , power consumption shall not exceed 250 W during load activation (see 4.8.4.3.1.4).

3.4.3.2 Output power.

3.4.3.2.1 Extinguisher activation power. The MSEC shall deliver extinguisher activation power (drive signals) in accordance with signal B of table II. The duration of these signals shall be 33 to 38 ms. The HSEC shall provide two extinguisher activation signals simultaneously. During all periods of quiescent system operation, current shall not exceed 5 mA at 16 to 30 V dc through each of the solenoid continuity and pressure/flow circuits (see 4.8.4.3.2.1).

3.4.3.2.2 Fire warning circuit.

3.4.3.2.2.1 Fire warning (small fire). The fire warning circuit shall be energized only when a fire equal to or greater than the small fire early warning threshold is attained. The fire warning lamp signal (see table VI) shall not latch (see 4.8.4.3.2.2).

3.4.3.2.2.2 Audible fire warning circuit. The MSEC shall provide an audible fire warning activation signal (see table VI) whenever a fire warning circuit becomes energized by either a small or large fire signal (see 4,8.4.3.2.2).

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3.4.3.2.3 Extinguisher status. The MSEC shall signal the test and alarm panel on the availability of each extinguisher A 16 to 30 V dc signal (see table VI) shall be output on the MSEC connector J1 for the extinguisher status lamps whenever an open circuit exists at MSEC connector J2 for the pressure/flow or solenoid circuits of that extinguisher (see table V). The signal shall remain so long as an open circuit condition exists (see 4.8.4.3.2.3 and 6.7.8).

3.4.3.2.4 BITE. The BITE system (see 3.3.5) shall operate on an intermittent (press and release to initiate test) basis. If no fault is found, a pass signal, which requires not more than 2 W, shall be generated for 4 to 6 s. If a fault is found, a fail signal shall be generated which requires not more than 2 W continuing until a retest indicates the fault has been corrected (see 4.8.4.3.1.4).

3.4.3.2.5 Short circuit of OFSA power. The MSEC shall be capable of withstanding a short of indefinite duration between the power output pin and return pin to any OFSA. Such a short shall not cause *a* false alarm, and the MSEC and remaining OFSA'S shall remain operational. Upon removal of the short, the system shall immediately regain full operational capability (see 4.8.4.3.2.4).

3.4.3.2.6 Electrical-manual release (extinguisher activation power). A manual input to the electrical-manual release, located on the test and alarm panel or similar device, shall activate two extinguisher's. The electric power to activate the extinguishers shall be drawn through the MSEC of the system. Application of the manual release signal to the appropriate pin in MSEC connector J1 as specified in table VI shall result in output signals on applicable extinguisher activation pins of MSEC connector J2 (see table V) (see 4.8.4.3.2.5 and 6.7.10).

3.4.3.2.7 Ventilation fan. The MSEC shall switch the ventilation fan on in not less than 8 s and not more than 10 s after the discharge of the fire extinguishers (see 4.8.4.3.2.6).

3.4.4 Electromagnetic interference (EMI). The system shall comply with the requirements of MIL-STD-461. The system shall not false alarm. The system shall not cause any EMI problems for other vehicle components. Where required, pins shall be shielded per MIL-STD-461 (see 4.8.4.4).

3.4.5 Nuclear hardening. Requirements for nuclear survivability shall be as specified in a separate classified (confidential) document entitled "Nuclear Survivability Criteria for the Automatic Optical Fire Detection and Suppression Systems for Combat Vehicles" (U) (see 4.8.4.5 and 6.9).

3.5 Environmental. The system shall ❑eet the requirements of 3.3.5 after and when specified during, exposure to the environmental conditions specified below. The system shall not false alarm (see 4.8.5).

3.5.1 Temperature. The system shall meet the requirements of 3.5 during exposure to the temperature extremes described below (see 4.8.5.1).

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3.5.1.1 High temperature. The MSEC and extinguishers shall operate at temperatures up to +71°C (+160°F) for extended periods of time. The OFSA'S shall operate at temperatures up to +125°C (+257″F) for extended periods of time (see 4.8.5.1.1).

3.5.1.2 Low temperature. The system shall operate at temperatures as low as -51°C (-60°F) for extended periods of time (see 4.8.5.1.2).

3.5.1.3 Temperature shock. The system shall withstand rapid changes in temperature between temperature extremes of -51 and +71°C (-60 and +160°F) (see 4.8.5.1.3).

3.5.2 Vibration. The system shall meet the requirements of 3.5 during vibration at the amplitudes and frequencies shown in figure 6 in each of three mutually perpendicular axes (see 4.8.5.2).

3.5.3 Shock. The system shall meet the requirements of 3.5 during exposure to sawtooth shocks as described in figure 7, applied in both directions of three mutually perpendicular axes (see 4.8.5.3).

3.5.4 Leakage. The system components shall meet the requirements of 3.5 and shall not evidence significant leakage after immersion in liquids as follows. The pre-test and post test weight of each component shall be within 0.5 grams (0.02 ounces) of each other (see 4.8.5.4).

a. Immersion in water under air pressure of 12 kilopascal (kPa) [1.7 pounds per square inch (psi)] (see 6.7.6).

b. Immersion in diesel fuel, conforming to DF-2 of VV-F-800, to a depth of 50 mm (2 in).

3.5.4.1 Water jet. When subjected to a water-jet and wiped dry, the components of the system shall not evidence leakage, and the windows of the OFSA shall not be broken or damaged. Subsequent to spraying the system shall meet the requirements of 3.5 (see 4.8.5.4.1).

3.5.5 Salt fog. The system shall meet the requirements of 3.5 and show no evidence of corrosion after prolonged exposure to a salt laden atmosphere (see 4.8.5.5).

3.5.6 Fungus. The system shall meet the requirements of 3.5 and shall evidence no microbiological growth following prolonged exposure to an environment favorable to fungus growth (see 4.8.5.6).

3.5.7 Sand and dust. The system shall meet the requirements of 3.5 following exposure to air driven sand and dust (see 4.8.5.7).

3.5.8 Humidity. The system shall meet the requirements of 3.5 during operation in high humidity environments (see 4,8.5.8).

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3.6 Finish. All component exterior surfaces, except lens windows, mounting surfaces, and electrical connections, shall be painted in conformance with requirements of the applicable part drawings (see 4.8.2).

3.7 Identification nameplate and markings.

3.7.1 Nameplate. Each component of the system shall have a nameplate attached in a conspicuous location. The nameplate shall have the following minimum information (see 4.8.2):

Component Name

National Stock Number (NSN)

Serial Number

Manufacturer's Name or Identification Code

3.7.2 Marking. The components of the fire system shall be marked in accordance with the applicable drawing and MIL-STD-130 (see 4.8.2).

3.7.2.1 Extinguishers. The extinguishers shall be marked in accordance with MIL-V-62547 (see 4.8.2).

3.8 Workmanship. Workmanship shall assure a product free of burrs, scratches, sharp edges, chips, corrosion, scale, and dirt, and shall melt requirement 9 of MIL-STD-454 (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements .

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2.1 Test equipment and standard conditions. Use of equipment, equipment designs and procedures other than as specified herein shall be subject to approval of the contracting officer,

4.1.2.2 Special test equipment. Special test equipment as well as commercial test equipment is required to perform some of the qualification tests.

4. 1.2.2.1 Equipment for response characteristics test. Standard or commercial test apparatus required for the response characteristics (small and large fires) tests are given in table VII.

TABLE VII. Components.

Description	Model number and manufacturer
Sixty-two millimeter circular shutter; normally closed; with light emitting diode at the full-open position acting as a synchronization switch	Uniblitz Model 262X2H25X5; Vincents Associates; Rochester, NY or equivalent
Shutter control unit (shutter timing and drive unit)	Uniblitz Model SD-1000; Vincents Associates; Rochester, NY or equivalent
Mounting bracket	Model 55529; Electro Corporation Birmingham, MI or equivalent
Solid state - DI - PROX Proximity switch, 12 V dc	Model 55525; Electro Corporation Birmingham, MI or equivalent

4.1.2.2.2 Use of special equipment indicated in figures 8, 9 and 10.

a. Test equipment specified in 4. 1.2.2.1 shall be used to perform the tests described in 4.8.4.1 and illustrated in figures 8, 9 and 10. These tests are to determine if the system responds to a fire stimuli and the approximate (+ 1 s) time frame. The sheet metal cover over the front of the-shutter assembly (the cover furthest from the shutter leafs) shall be removed to obtain a 90 degree (°) field of view (FOV) for the OFSA during the tests. The shutter assembly and the OFSA under test shall

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be mounted to the same base plate, so when the shutter assembly with base plate is rotated about a fixed pivot to obtain the 90° FOV, the OFSA automatically shall rotate the same amount. The OFSA shall then be mounted at right angles to its original position with the optical window facing the same direction (the preceding shall be repeated to obtain the FOV in this second axis). An alternate method is shown in figure 9. Instrumentation for either setup is not required. A stop-watch is sufficient. If instrumentation is used, figure 10 provides a schematic diagram of the instrumentation, required auxiliary items, and the system. The shutter control unit shall be programmed for a maximum of 5 s of OFSA exposure to the fire stimuli.

b. The MSEC output to the test and alarm panel and extinguishers shall be measured and shall be recorded on the oscillograph, with the oscillograph being optional. When the shutter controller is triggered, time zero or the starting time shall be recorded.

4.1.2.3 Apparatus. In addition to standard measuring instruments (see table VIII), test apparatus shall include chambers for conducting low temperature, high temperature, sand and dust, fungus, humidity, and salt fog tests (see 4.8.5).

TABLE VIII. Measuring instrument tolerance.

Measuring instrument	Accuracy
Voltmeter dc	± 0.25%
Ammeter dc	± 0.50%
Milliammeter	+ 0.50%
Temperature indicating system	± 4.5°C (± 8°F)
Timing/counter	+ 1.0%
Oscilloscope (four *or* six channel, time measuring)	+ 3.0%
Oscillograph (multi-channel)	+ 1.0%
Tachometer - generator	+ 1.5%
Time measuring system	± 1.0%
Weighing system	+ 0.25 gram at 150 to 1500 grams. (+ 0.009 ounces at 5 to 53 ounces).

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4.1.2.4 Calibration of test instruments. Standard instrument tolerance accuracies are listed in table VIII. Records of calibration of test instruments at intervals sufficient to establish the required accuracy shall be made available to the Government for review. The inspector may refuse to conduct an inspection where accuracy of test instruments have not been established to be in accordance with MIL-STD-45662. Where compliance with MIL-STD-45662 is not possible, special calibration requirements shall be prepared for approval by the Government.

4.1.3 Qualified products. When a part or component is specified to conform to a specification having a QPL, the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order and quantity.

4.2 Classification of inspection:

a. Qualification inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 + 10°C (73 + 18°F)
b. Barometric pressure 725 + 50 millimeters of mercury (mm Hg)
   - 75
   [28.5 +2 inches of mercury (in Hg)]
   -3
c. Relative humidity 50 + 30%

4.3.1 Voltage setting. For all tests, the input voltage to the system shall be supplied by batteries (22 to 28 V de), or a variable power supply of 60 A current capacity, as specified herein or as appropriate.

4.3.2 Test temperature stabilization. Tests shall be conducted after the test sample has been allowed sufficient time for temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary in temperature more than 2°C (3.6°F) per hour.

4.4 Qualification inspection. Qualification inspection shall consist of all the examinations and tests of this specification, as specified in table IX (see 6.3).

404.1 Qualification samples. Qualification samples shall consist of four fire suppression systems. Unless otherwise specified (see 6.2), a system shall consist of four OFSA, one MSEC and four extinguishers (see 6.4). For type I systems, an additional 12 OFSA'S and 3 MSEC'S shall be submitted

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for testing for conformance to 3.4.1.4. The qualification samples produced with equipment and procedures used in production, shall be forwarded to a test facility set forth in the letter of authorization to submit samples. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-S-62544(AT) and number under authorization (reference authorizing letter and number).

TABLE IX. Classification of inspections.

Title	Requirement	Inspection	Qualification sample				Quality Conformance Examination	Control
			1	2	3	4		
Materials and construction	3.2 thru 3.2.2 and 3.3 thru 3.3.2	4.8.1	X					
Defects (see table X)	3.3, 3.3.2 and 3.6 thru 3.8	4.8.2	X	X	X	X	X	
Features	3.3.3 thru 3.3.6	4.8.3	X	X				X
Performance	3.4	4.8.4						
Response characteristics	3.4.1	4.8.4.1						
Early fire warning (small fire)	3.4.1.1	4.8.4.1.1	X	X	X	X		X
Secondary sensitivity threshold (large fire)	3.4.1.2	4.8.4.1.2	X	X	X	X		X
Response time	3.4.1.3	4.8.4.1.3	X	X	X	X		X
Penetration response time (type I system only)	3.4.1.4	4.8.4.1.4	X	X	X	X		
Start-up	3.4.2	4.8.4.2		X	X			X
Input and output power	3.4.3	4.8.4.3						
Input power	3.4.3.1	4.8.4.3.1						
Quiescent condition	3.4.3.1.1	4.8.4.3.1.1	X	X	X	X		X
Input voltage decay	3.4.3.1.2	4.8.4.3.1.2			X	X		X

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TABLE IX. Classification of inspections - Continued.

Title	Requirement	Inspection	Qualification sample				Quality Conformance Examination	Control
			1	2	3	4		
Steady state voltage extremes	3.4.3.1.3	4.8.4.3.1.3			X	X		X
BITE power	3.4.3.1.4	4.8.4.3.1.4	X	X				X
Output power	3.4.3.2	4.8.4.3.2						
Extinguisher activation power	3.4.3.2.1	4.8.4.3.2.1	X	X				X
Fire warning circuit	3.4.3.2.2	4.8.4.3.2.2						
Fire warning (small fire)	3.4.3.2.2.1	4.8.4.3.2.2	X	X				X
Audible fire warning circuit	3.4.3.2.2.2	4.8.4.3.2.2	X	X				X
Extinguisher status	3.4.3.2.3	4.8.4.3.2.3	X	X				X
BITE	3.4.3.2.4	4.8.4.3.1.4	X	X				X
Short circuit of OFSA power	3.4.3.2.5	4.8.4.3.2.4	X	X				
Electrical-manual release	3.4.3.2.6	4.8.4.3.2.5	X	X				X
Ventilation fan	3.4.3.2.7	4.8.4.3.2.6	X	X				X
EMI	3.4.4	4.8.4.4			X	X		
Nuclear hardening	3.4.5	4.8.4.5		X	X			X
Environmental	3.5	4.8.5	X	X				
Temperature	3.5.1	4.8.5.1	X	X				
High temperature	3.5.1.1	4.8.5.1.1	X	X				
Low temperature	3.5.1.2	4.8.5.1.2	X	X				
Temperature shock	3.5.1.3	4.8.5.1.3	X	X				
Vibration	3.5.2	4.8.5.2	X	X				
Shock	3.5.3	4.8.5.3	X	X				
Leakage	3.5.4	4.8.5.4	X	X				
Water jet	3.5.4.1	4.8.5.4.1	X	X				
Salt fog	3.5.5	4.8.5.5	X	X				
Fungus	3.5.6	4.8.5.6	X	X				
Sand and dust	3.5.7	4.8.5.7	X	X				
Humidity	3.5.8	4.8.5.8	X	X				

4.4.2 Qualification inspection failure. Failure of any sample item to pass the examinations and tests of table IX shall be cause for refusal to grant qualification.

4.5 Quality conformance inspection.

4.5.1 Sampling.

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4.5.1.1 Lot formation. An inspection lot shall consist of systems of one type and class, from an identifiable production period, from one manufacturer, submitted at one time for acceptance (see 6.4).

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table X.

TABLE X. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Incorrect dimensions affecting interchangeability (see 3.3).	SIE 1/
102	Damaged [cracked, scratched, broken OFSA window(s)] (see 3.3).	Visual
103	Damaged connectors (see 3.3.2).	Visual
104	Improper nameplate (see 3.7.1).	visual
105	Improper identification markings (see 3.7.2)	Visual
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3).	SIE
202	Improper finish (see 3.6).	Visual
203	Faulty workmanship (see 3.8).	Visual

1/ SIE = Standard inspection Equipment.

4.6 Control tests. Control tests shall be conducted on 1 system from each lot of 250 units consecutively produced, except that not more than 2 systems shall be tested in a l-month period. The system shall be subjected to the control tests specified in table IX.

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4.7 Failure. Failure of any system to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.2 and 3.3 through 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.2 and 3.6 through 3.8 shall be determined by examination for the defects listed in table X. Examination shall be visual or by measurement with SIE.

4.8.3 Features. Conformance to 3.3.3 through 3.3.6 shall be determined by exercising the system components, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.8.4 Performance.

4.8.4.1 Response characteristics.

4.8.4. 1.1 Early fire warning (small fire). To determine conformance to 3.4.1.1, the system shall energize the fire warning lamp and audible fire warning signal pins in not more than 5 s in response to a 130-mm (5-in) diameter pan-fire, 76 mm (3 in) minimum depth containing 840 cm^3 (51 in^3) of diesel fuel conforming to DF-2 of VV-F-800 at a distance of 1200 mm (48 in) from the OFSA, without producing an extinguisher drive signal. The fire warning circuit shall not lock in the energized position (see table VI). Time and MSEC outputs shall be recorded. The tests cited above shall be conducted as shown on figures 8, 9, and 10.

4.8.4. 1.2 Secondary sensitivity threshold (large fire). To determine conformance to 3.4.1.2, the procedure of 4.8.4.1.1 shall be repeated except that the pan-fire shall be 380 mm (15 in) from the OFSA. The system shall respond as specified in 3.4.1.2.

4.8.4. 1.3 Response time. To determine conformance to 3.4.1.3, the valve upon receiving an activation signal shall begin to discharge the Halon within 10 ms. The valve shall continue to discharge the liquid phase of Halon within the time frames specified for class 1, 2 and 3 extinguishers.

4.8.4. 1.4 Penetration response time (type I system only). Conformance to 3.4.1.4 shall be determined in accordance with 4.8.4. 1.4.1 through 4.8.4.1.4.5.

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4.8.4.1.4.1 Fabricated target enclosures. Tests shall be performed on both steel and aluminum armor targets to assure interchangeability. The two different material targets shall be used in the test described in 4.8.4.1.4.5. The contractor shall design and fabricate a reuseable target enclosure suitable for the accomplishment of the tests. The contractor shall obtain Government approval for the design of the enclosure. The enclosures shall include the following:

a. The enclosure shall be rectangular in shape and of a configuration having an interior void volume of 13 + 0.3 cubic meters (m^3) [472 + 12 cubic feet (ft^3)], which is the void volumetric size of—the M60 tank, less the engine compartment. The interior of the enclosure and the ballistic test plate shall be painted using enamel conforming to TT-E-489, type 1, class A, color number 17875.

b. It shall be reusable with only minimum repair following each munition shot. The interior paint shall be touched-up as necessary between shots to assure a uniform white interior.

c* It shall contain a square projectile entrance opening in one side of the enclosure and a square exit opening in the other side directly opposite the entrance opening.

d. It shall have provisions for attaching entrance and exit aluminum and steel target plates. Entrance plates shall be attached to the outside of the enclosure. Aluminum plates shall be 32 mm (1 1/4 in) thick and steel plates 50 mm (2 in) thick. Three steel plates will be layered to achieve a 150-mm (6-in) thickness. The exit plates are to be square, mounted on the inside of the enclosure and of the same thickness as the entrance targets. Plates shall be removable between shots.

e. It shall incorporate a removable 38-liter (10-gallon) (minimum) fuel tank fabricated of 5-mm (3/16-in) thick aluminum plate. The tank shall be located between the penetration entrance and exit target, at a distance of not less than 1000 mm (40 in) and not more than 1500 mm (60 in) from the point of munition penetration. A method for securely mounting the fuel tank shall be provided. The fuel tank shall have provision for filling with diesel fuel.

f. It shall incorporate a removable 7.5-liter (8-quart) aluminum cylinder to be filled with fire retardant hydraulic fluid and pressurized with 4826 kPa (700 psi) nitrogen to simulate an M60 accumulator. This cylinder shall be located and mounted as specified for the fuel tank. The contractor should consider using an aluminum fire extinguisher cylinder of approximately comparable volume. Cylinders shall be rated for 12 410 kPa (1800 psi) minumum service pressure.

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g. It shall incorporate on one end a panel for mounting the 12 OFSA'S being tested. The panel shall allow for easy servicing and replacement of the OFSA'S between tests. The OFSA'S shall be no closer than 380 mm (15 in) to any part of the fuel tank or accumulator, or further than 2100 mm (84 in) from the point of munition penetration.

h. It shall incorporate an electrically operated semi-automatic fire extinguisher(s) remotely controlled with a 2 to 3 time overkill capability. There shall be no deterrent to extinguishant flow into the enclosure such as an installed anti-recoil device(s) in the fire extinguisher(s)*

i. It shall incorporate a personnel access door for installing exit plates and servicing equipment. It shall also contain, on the exit target side, one or more spring loaded hatch door(s) to relieve any excessive over pressurizing. Consideration should be given to installing a channel for ease in sliding the exit plates into the enclosure.

j. It shall provide for the attachment of the necessary instrumentation to monitor tests.

4.8.4.1.4.2 Instrumentation. MSEC'S and required instrumentation shall, to the extent possible, be located outside the target enclosures to minimize damage . Instrumentation and equipment required for response time tests shall be protected by armor from fragments generated by the penetration. Tests for response time to hydrocarbon fire ignited subsequent to and resulting from penetration of armor shall be conducted with instrumentation and equipment capable of determining the start of munition penetration, time when the fire is ignited, and response time for OFSA'S and MSEC'S. Four OFSA'S instrumented with an analog output shall also be utilized to confirm the discriminatory capabilities of the test samples.

4.8.4.1.4.3 Armor. The aluminum target plates, entrance and exit, required for test described in 4.8.4. 1.4.5 shall be 32 mm (1 1/4 in) thick of either 5083 or 7039 material. The steel target plates, entrance and exit shall be 50-mm (2-in) thick ballistic steel material. Additional layers of the 50-mm (2-in) plate shall be used to meet the 150-mm (6-in) thickness requirements.

4.8.4.1.4.4 Munitions. The following munition sizes have been chosen for test purposes: 38-mm (1.5-in), 90-mm (3.5-in), 105-mm (4-in), and 120-mm (5-in) chemical energy. Specific model designations and velocities for the munitions will be determined, based upon the type and thickness of armor materials used in the tests and munitions availability.

4.8.4.1.4.5 Penetration response. One round of each of the 38-mm (1.5-in) and 105-mm (4-in) chemical energy munitions shall be shot through the 32-mm (1 1/4-in) thick aluminum entrance armor target; then through not less than 1000 mm (40 in) of air, and then through the fuel tank containing a

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hydrocarbon fuel (diesel fuel DF-2 per W-F-800) preheated to sufficient volatility to assure a fire. The tests shall then be repeated with the simulated accumulator, containing preheated fire retardant hydraulic fluid conforming to MIL-H-46170, replacing the fuel tank. These tests shall then be repeated using 50-mm (2-in) thick steel target plates with a fuel tank and then with an accumulator using 105-mm (4-in) chemical energy rounds. After each test, inoperative OFSA'S shall be replaced and new fuel tanks or accumulators, as applicable, installed. During these tests, one fire extinguisher system, consisting of four OFSA'S, one MSEC and one test and alarm panel (see 6.2) shall be operational. The remaining eight OFSA'S and two MSEC'S shall either be connected to extinguishers or valve simulators, and test and alarm panels or simulators. For each test, the response time from start of penetration to ignition of the fire to OFSA large and small fire signal output and to MSEC output signal shall be measured and recorded for each system. Any OFSA or MSEC not functioning properly or not responding within the times specified in 3.4.1.4, or the test and alarm panel not perperly operating, shall be considered a failure. The test and alarm panel shall be rotated to a different OFSA and MSEC system following each live fire test .

4.8.4.2 Start-up. To determine conformance to 3.4.2, a complete system (including discharge valves and a test and alarm panel) shall be set Up and allowed to stand in an inactive mode for 6 to 24 hours. An 18 V dc power source shall be switched-on in a step-function mode energizing the total system. The discharge valves shall not false alarm. The test shall be repeated using a 30 V dc power supply.

4.8.4.3 Input and output power. The procedures to conduct tests for the various input and output current draw requirements of 3.4.3 at different input voltages are described in the following paragraphs. The test setups are indicated in figures 11 and 12. Data collected during these tests shall be recorded.

4.8.4.3.1 Input power.

4.8.4.3.1.1 Quiescent condition. To determine conformance to 3.4.3.1.1, the system shall be connected to a variable power supply set at 24 V dc. Using the setup of figure 11, quiescent condition readings of milliamperes and voltage shall be taken.

4.8.4.3.1.2 Input voltage decay. Conformance to 3.4.3.1.2 shall be determined as follows. By changing the milliammeters to ammeters and inserting an oscilloscope or oscillograph into the valve circuit, the setup of figure 12 may be used to check the system for false alarms during voltage decays.. The variable power supply shall be capable of 60 A current draw, regardless of the voltage setting. Set the power supply to 30 V dc, with the total system in the quiescent mode. Energize the 1/2–revolution per minute (rpm) motor so that the power supply voltage decreases to zero. This can take from 1 to 15 minutes depending on the voltage selector knob sensitivity and the number of revolutions required to come to zero. Shut the motor off. This test shall be repeated three times starting at 30 V dc and ending at zero.

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4.8.4.3.1.3 Steady state voltage extremes. To determine conformance to 3.4.3.1.3, the system shall be connected to a variable power supply. Using the setup of figure 11, the voltage shall be set at 2.5 V dc for 3 minutes of system quiescent operation. It shall then be increased in 5.0 V dc increments for 3 minute intervals until reaching 37.5 V dc. It shall then be set at 40 V dc for the final 3 minutes.

4.8.4.3. 1.4 BITE power. Using the setup of figure 11, the BITE button on the test and alarm panel shall be pressed and released. A recording of current draw and voltage shall be made to determine conformance with requirements of 3.4.3.1.4 and 3.4.3.2.4. The test shall be repeated with all OFSA windows blocked to simulate dirty-window faults. The test shall then be repeated with optical lens or window facing directly into sun.

4.8.4.3.2 Output power.

4,8.4.3.2.1 Extinguisher activation power. To determine conformance to 3.4.3.2.1, using the setup of figure 11, current draw through each extinguisher activation circuit shall be measured at MSEC connector J2 and recorded during quiescent operation. An oscilloscope or oscillograph [to be operated at 800 mm/s (32 in/s)] shall be added to the setup of figure 11 for operational tests. All current and voltage readings shall be recorded (also see 4.8.4.3.2.2 for related readings if tests are run simultaneously). A movie light [625 W quartz DWY lamp (Sylvania S.G. -55 or equivalent)] shall be placed at a distance which shall assure activation of the system when the light is energized. The movie light shall be energized triggering the oscilloscope or oscillograph and activating the system. The test shall be repeated using the electrical-manual release of 3.4.3.2.6,

4.8.4.3.2.2 Fire warning circuit. When testing to the previous extinguisher activation power requirements, 4.8.4.3.2.1, the fire warning circuit responses (visual and audible), current drain and voltage required shall be recorded and shall meet the requirements of 3.4.3.2.2.1 and 3.4.3.2.2.2.

4.8.4.3.2.3 Extinguisher status. To determine conformance to 3.4.3.2.3, an extinguisher switch panel shall be prepared with switches installed to represent the pressure/flow and continuity switches associated with each extinguisher (total of eight required). These switches shall be appropriately wired to the MSEC to simulate the extinguisher switches (see table V). All switches shall be closed at the beginning of the test. Extinguisher status lamp signals for the four extinguishers shall be measured at the appropriate pins on MSEC connector J1 (see table VI). After system power is turned on, the pressure/flow switch for each extinguisher shall be sequentially opened. After closing all switches, the continuity switch for each extinguisher shall be sequentially opened. The test shall be repeated with both switches for each extinguisher being sequentially opened.

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4.8.4.3.2.4 Short circuit of OFSA power. To determine conformance to 3.4.3.2.5, power output and return pins of each OFSA shall be shorted together, one OFSA at a time, for at least a 2-minute duration. During each short period, another OFSA shall be stimulated by a large fire radiation source.

4.8.4.3.2.5 Electrical-manual release (extinguisher activation power). To determine conformance to 3.4.3.2.6, the electrical-manual release shall be tested according to 4.8.4.3.2.1.

4.8.4.3.2.6 Ventilation fan. To determine conformance to 3.4.3.2.7, the MSEC shall be instrumented to establish the time duration between an extinguisher drive signal and output of a ventilation fan activation signal on the appropriate pin of MSEC connector J1 (see table VI). The test shall be repeated for the second shot. The signal and duration shall be as specified in table VI. The ventilation fan tests may be conducted in conjunction with the response characteristics tests of 4.8.4.1.

4.8.4.4 EMI. To determine conformance to 3.4.4, the system shall be subjected to the test procedures specified in MIL-STD-462 and TACOM Report No. 12679. The system shall not false alarm or cause any EMI problem for other vehicle components.

4.8.4.5 Nuclear hardening (see 6.9).

4.8.4.5.1 Qualification. Conformance to 3.4.5 shall be determined on the basis of circuit analysis and laboratory testing.

4.8.4.5.2 Control. Conformance to 3.4.5 shall be determined by inspection of contractor records providing certification that the systems have been fabricated in accordance with the design verified and approved for qualification.

4.8.5 Environmental. Unless otherwise specified (see 6.2), the system shall be tested as specified in 4.8.5.1 through 4.8.5.8. The system shall be operated in the quiescent mode (stand-by) except as otherwise specified herein. After each environmental exposure, the system shall undergo a complete system check in accordance with 3.3.5.

4.8.5.1 Temperature. The system shall be tested as specified in 4.8.5 during exposure to the temperature extremes specified below (see 3.5.1).

4.8.5. 1.1 High temperature. To determine conformance to 3.5.1.1, high temperature testing requires two ovens. The OFSA'S shall be in one oven and the temperature shall be maintained at +125°C (+257"F). The MSEC and the extinguishers shall be in the second oven and the temperature maintained at +71°c (+160°F). The system shall be interconnected by a harness assembly, a portion of which shall be outside of both ovens at existing ambient temperature. The test and alarm panel simulator shall be an active part of the circuit and also may be at ambient temperature. A 22 to 28 V dc battery, external to the ovens, may be connected to the test and alarm panel simulator

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or the MSEC and be capable of driving (through the MSEC) two extinguishers simultaneously at no less than 20 A. The two ovens shall be maintained at their respective high temperatures for a period of 240 + 24 hours. There shall be no false alarms. A recorder, with not less than four channels, shall measure and record voltage across each extinguisher continuously during the test for indications of any extinguisher false alarms. The samples shall be inspected at least two times per working day, a minimum of 2 hours apart, and the inspection shall be noted on the recorder's trace. During these inspections, the extinguishers shall be activated by exposing the OFSA'S to an appropriate radiation source. Also, two extinguishers shall be electrically actuated and two extinguishers shall be mechanical-manually actuated. These operations shall be noted on the trace.

4.8.5.1.2 Low temperature. To determine conformance to 3.5.1.2, the system shall be placed in a low temperature chamber. The instrumentation requirements shall be identical to the preceding paragraph as to type(s) and location(s). All the test conditions for high temperature testing (see 4.8.5.1,1) shall apply to low temperature testinng except that the components shall be in the same test chamber and the temperature shall be lowered and maintained at -51°C (-60″F). The test samples shall be allowed to thermally stabilize for 24 hours and then soak for an additional 72 hours.

4.8.5. 1.3 Temperature shock. To determine conformance to 3.5.1.3, test samples shall be put through three high temperature to low temperature cycles as described below. The instrumentation requirements shall be identical to the preceding paragraph as to type(s) and location(s). The components shall be in the same test chambers. The system shall be set up and checked with a radiation source which shall activate the valve simulators. The samples shall be exposed to ambient temperature for no more than 5 minutes during transfer. The procedures of 4.8.5.1.1 and 4.8.5. 1.2 apply except:

Step 1. The MSEC, OFSA'S, extinguishers and appropriate cables and harness assemblies shall be placed in an oven preheated to +71°C (+160°F) and maintained for 24 ± 2 hours. While at this high temperature, the OFSA'S shall activate the extinguishers when exposed to an appropriate radiation source.

Step 2. Immediately following step 1 the test sample shall be transferred to a precooled low temperature chamber and maintained at -51°C (-60″F) for a period of 24 + 2 hours. While at this low temperature the OFSA'S shall activate the extinguishers when exposed to an appropriate radiation source.

Step 3. Repeat step 1.

Step 4. Repeat step 2.

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Step 5. Repeat step 1.

Step 6. Repeat step 2. Twenty minutes after transfer, the requirements of 3.3.5 shall be met. At the end of step 6 the test sample shall be returned to ambient temperature.

4.8.5.2 Vibration. To determine conformance to 3.5.2, the test samples shall be mounted by normal mounting provisions to a vibration exciter capable of producing and of transmitting the specified vibrations. The samples shall be connected as in use and also connected to the instrumentation necessary for this test. The vibration acceleration level shall be in accordance with figure 6, applied in 15 minute sweep times in accordance with figure 13. Resonant frequencies of the sample shall be conducted for the four most severe resonent frequencies of the samples. Resonant search and a dwell of 30 minutes at each resonance shall be part of the total maximum cycle time of 3 hours. The frequency of applied vibration shall be swept over the specified range in accordance with the ascending and descending curve of figure 13. Regardless of the axis in which the MSEC Is being vibrated, there shall be at least one OFSA mounted in each orthogonal axis. Graphic records shall be maintained by the contractor to insure that the three axes vibration requirements of the OFSA'S are met for every change in direction of vibration of MSEC. The system shall pass the tests specified in 4.8.5 during exposure.

4.8.5.3 Shock. To determine conformance to 3.5.3, test samples shall be exposed to the shock pulse defined in figure 7 in each direction of three mutually perpendicular axes. Regardless of the shock direction of the MSEC in each of the orthogonal axes, there shall be at least one OFSA in each orthogonal axis (3 OFSA'S at a time). Graphic records shall maintained by the contractor to insure that the two directions of shock for the three axes test requirements of the OFSA'S are met for every change in direction of shock of the MSEC. The system shall pass the tests specified in 4.8.5 during exposure.

4.8.5.4 Leakage. To determine conformance to 3.5.4, system components shall not evidence any leakage when submerged in liquids as specified in a and b below. These components shall not be operated during these tests. Subsequent to these tests, the system shall pass the tests specified in 4.8.5.

a. Water. The temperature of both the water and test sample shall be +23 ± 10°C (+73 ± 18°F). The test sample shall be completely immersed in water 80 that the upper most part is 50 ± 25 mm (2 ± 1 in) below the surface of the liquid. The pressure of the air above the liquid shall be reduced to 12 kPa (1.7 psi) and maintained for 1 minute or until air bubbles substantially cease to be given off by the liquid, which ever is the longer time. The air pressure above the liquid shall then be increased to 15 kPa (2.2 psi) and maintained for 60 minutes. Bubbles coming from within the component shall be considered leakage; however bubbles which result from trapped air on the various exterior surfaces of the test sample shall not be considered a leak.

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b. Diesel fuel. Prior to immersion in diesel fuel conforming to DF-2 of VV-F-800, the temperature of the test samples shall be +45° + 3°C (+113 + 5°F); the temperature of the diesel fuel shall be +18 + 5°C (+64 + 9°F). The container shall be of sufficient capacity so that the immersion of the test sample shall not raise the temperature of the fuel more than 3°C (5°F). Each test sample shall remain immersed for 120 + 5 minutes. The upper most point of the test item shall be 50 mm (2 in) below the surface. Upon completion of the test period remove the components from the oil and wipe the exterior surfaces only. The components shall be weighed before and after this test. The weights shall not vary more than 0.5 gram (0.02 ounce).

4.8.5.4.1 Water jet. To determine conformance to 3.5.4.1, the system shall be cleaned by water jet spray. The jet spray shall be applied at right angles to the surface being cleaned at a distance of not less than 300 mm (12 in) from that surface to the jet. The surface shall be cleaned at not less than 0.02 square meter per minute (0.2 square feet per minute). The water jet shall be derived from a nozzle having a maximum orifice diameter of 6 mm (15/64 in) and a nozzle pressure of 345 + 105 kPa (50 + 15 psi). The system shall subsequently pass the tests specified in 4.8.5.

4.8.5.5 Salt fog. To determine conformance to 3.5.5, a non-operating system shall be subjected to tests of MIL-STD-810, method 509 except the test shall last 200 hours. The system shall subsequently pass the tests specified in 4.8.5.

4.8.5.6 Fungus. To determine conformance to 3.5.6, a non-operating system shall be subjected to fungal incubation as specified in MIL-STD-810, method 508 for a period of 90 days. The system shall subsequently pass the tests specified in 4.8.5 and evidence no biological growth.

4.8.5.7 Sand and dust. To determine conformance to 3.5.7, a non-operating system with connectors and harnesses attached shall be exposed to blowing fine sand as specified below. During this test, the OFSA shall be rotated so that the windows and connectors receive equal blast time. The MSEC shall be positioned in such away that the connectors receive the blasts. The system shall subsequently pass the tests specified in 4.8.5.

a. Apparatus. The test facility shall consist of a chamber and accessories to control dust concentration, velocity, temperature and humidity of dust laden air. In order to provide adequate circulation of the dust laden air, no more than 50% of the cross-sectional area (normal to air flow) and 30% of the volume of chamber shall be occupied by the test item(s). The chamber shall be provided with a suitable means of maintaining and verifying the dust concentration in circulation. A minimum

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acceptable means for doing this is by use of a properly calibrated smoke meter and standard light source. The dust laden air shall be introduced into the test space in such a manner as to allow it to become approximately laminar in flow before it strikes the test item.

b. Dust requirements. The dust used in this test shall be a fine sand [97 *to* 99% by weight silicon dioxide (SiO_2)] of angular structure, and shall have the following size distribution as determined by weight, using the US standard sieve series 2/:

1. 100% of this dust shall pass through a 100-mesh screen.
2. 98 ± 2% of the dust shall pass through a 140-mesh screen.
3. 90 ± 2% of the dust shall pass through a 200-mesh screen.
4. 75 ± 2% of the dust shall pass through a 325-mesh screen.

c. Procedure. Position the test item as near to the center of the chamber as practicable. If more than one item is being tested, there shall be a minimum clearance of 105 mm (4 in) between surfaces of test items or any other material or object capable of furnishing protection. Also, no surface of the test item shall be closer than 105 mm (4 in) from any wall of the test chamber. Orient the item so as to expose the most critical or vulnerable parts to the dust stream. The test item orientation may be changed during the test if so required.

Step 1. Set the chamber controls to maintain an internal chamber temperature of +23 ± 1.4°C (+73 ± 2.5°F) and a relative humidity of less that 22%. Adjust the air velocity to 533 + 76 meters per minute (m/min) [1,750 + 250 feet per minute (ft/min)]. Adjust the dust feeder to control the dust concentration at 0.3 ± 0.2 grams (0.01 ± 0.007 ounces) per cubic foot. Maintain these conditions for 6 hours.

Step 2. Stop the dust feed and reduce the air velocity to (91 + 61 m/min) (300 ± 200 ft/min). Raise the internal chamber-air temperature to +63 ± 1.4°C (+145 ± 2.5°F). Hold these conditions 16 hours.

Step 3. While holding chamber temperature at +63 ± 1.4°C (+145 ± 2.5°F) adjust the air velocity to 533 ± 76 m/min (1,750 + 250 ft/min). Adjust the dust feeder to control the dust concentration at 0.3 ± 0.2 grams (0.01 ± 0.007 ounces) per cubic foot. Maintain these conditions for 6 hours.

2/ 140-mesh silica flour as produced by the Ottawa Silica Company, Ottawa, Illinois, or equal, is satisfactory for use in the performance of these tests.

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Step 4. Turn off all chamber controls and allow the test item to return to standard ambient conditions. Remove accumulated dust from the test item by brushing, wiping, or shaking, care being taken to avoid introduction of additional dust into the test item. Dust shall not be removed by either air blast or vacuum cleaning.

Step 5. Operate and inspect the test item.

Step 6. Examine the system carefully for the presence of dust deposits which may affect performance.

4.8.5.8 Humidity. To determine conformance to 3.5.8, the system shall pass the tests specified in 4.8.5 during and after exposure to warm, high relative humidity in accordance with the applicable conditioning for ground electronic shall equipment of MIL-STD-810, method 507. After conditioning, the system shall be exposed to five continuous 48 hour cycles in accordance with figure 14. Prior to post exposure operation, the system shall be conditioned for 24 hours at +23 ± 3°C (+73 ± 5°F) and 50 + 10% relative humidity.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation , packaging, packing and marking for the desired level snail be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The systems are used to detect fires in military ground combat and tactical vehicles [such as the M992 Field Artillery Ammunition Support Vehicle (FAASV)]. The systems respond to optical radiation from exploding atomized or vaporized hydrocarbons and energizes the fire extinguishing system within 5 to 6 ms, depending on OFSA type, when an energy level equal to or greater than the large fire threshold (see 3.4.1.2) is attained. The systems are intended for use in personnel areas, and other compartments such as engine spaces. These systems are designed to protect against the potential danger of detonations, deflagration, and slow growth fires due to the presence of highly combustible fuels or other liquid hydrocarbons or flammable debris.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type and class of system (see 1.2 and 4.4).
c. Quantity of OFSA'S and quantity of extinguishers required per system (see 1.2 and 4.4).
d. The number of shots and number of extinguishers per shot, if other than as specified (see 3.3).
e. Specify interconnecting cable(s) and alarm panel to be used (see 3.3 and 4.8.4.l.4.5).

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f. If dip or siphon tubes are required (see 3.3.1.4.4)0
g. Responsibility for inspection, if other than as specified (see 4.1).
h. Responsibility for inspection equipment, if other than as specified (see 4.1.2).
i. Inspection conditions, if other than as specified (see 4.3).
j. If system shall be other than as specified for qualification inspection (see 4.4.1).
k. If environmental testing is not required (see 4.8.5 and 6.6).
1. Selection of applicable level and packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable QPL (QPL-62544), whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered in this specification. The activity responsible for this QPL is US Army Tank-Automotive Command, ATTN: AMSTA-GDS Warren MI 48397-5000 (see 3.1).

6.3.1 Retention of qualification. The retention of qualification shall consist of periodic verification to determine compliance of the quailfied fire suppression system with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the QPL and shall be included in the Notice of Qualification letter.

6.4 Fire suppression system sample. Each class furnished will be considered as individual samples (see 4.4.1 and 4.5.1.1).

6.5 Supersession data. This military specification supersedes ATPD-2070, revision A, 29 February 1984, Sensor, Fire, Optical: System with Amplifier, Standard Control, Electronic, and ATPD 2071, revision A, 29 February 1984, Extinguisher, Fire, Bromotrifluoromethane (Halon 1301): Fixed, for Automatic Systems, to the extent of its application to complete fire suppression systems.

6.6 Environmental. The system shall be capable of meeting the environmental requirements specified herein. At the option of the qualifying activity it may not be deemed necessary to re-test the system providing that all the major components have been obtained from sources specified on QPL's in effect at the time set for opening of bids (see 4.8.5 and 6.2).

6.7 Definitions.

6.7.1 Discriminatory system (type I). A discriminatory system is one that meets all the requirements of Section 3.

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6.7.2 Non-discriminatory system (type II). A nondiscriminatory system is described as one that shall meet the requirements of Section 3 except for 3.4.1.4.

6.7.3 False alarm. A false alarm exists when either an OFSA large or small fire signal or any HSEC drive signal is produced without proper input stimuli.

6.7.4 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.7.4.1).

6.7.4.1 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.7.5 Proximity switch. The units used here are solid state proximity switches which are integral units containing both sensing and switching elements, The unit senses and measures any metallic object at distances as giging of suppressant) of which a particular fire ‘
sensing and extinguishing system is capable. An extinguishing action consists of activating either one or two extinguishers.

6.7.8 Extinguisher status. Extinguisher status and extinguisher availability are used interchangeably.

6.7.9 Automatic. The extinguisher is activated, without any human decisions, by the system.

6.7.10 Electrical-manual. The extinguisher is activated by a human being who pushes an extinguisher activation button after receiving an alarm signal from the system or after seeing a fire not detected by the system.

6.7.11 Manual. The extinguisher is manually actuated by entirely mechanical means when there is an electrical malfunction or when electrical power is not available.

6.8 Laboratory safety precautions.

6.8.1 Halon 1301 exposure. Unnecessary exposure of personnel to either the natural agent or to the decomposition products (due to heat) of Halon 1301 shall be avoided. A ventilation blower shall be turned on, and doors and windows open during testing.

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6.8.2 Noise. During testing, hearing protection shall be worn at all times or the tests shall be conducted in an unoccupied noise insulated test chamber, cell or room.

6.8.3 Cold temperature affects. Direct or indirect contact with Halon 1301 extinguishing agent being discharged can cause frostbite burns. Personnel shall avoid positioning any part of the body in the agent discharge path.

6.8.4 Caution. The extinguisher can discharge if the antirecoil plug is not installed and the manual release is not locked prior to loosening the receiver bottle from its mounting. Such a discharge will turn the extinguisher into an uncontrolled projectile capable of serious personnel injury and property damage.

6.9 Nuclear survivability criteria. Vendors requesting the nuclear survivability criteria, which is classified confidential, shall possess the proper security clearance and a demonstrated need to know (see 2.1.2 and 3*4.5).

6.10 Subject term (key word) listing.

Brornotriflourornethane fire suppression system for combat ground vehicles
Fire suppression system for combat vehicles
HALON 1301 fire suppression systems for combat vehicles

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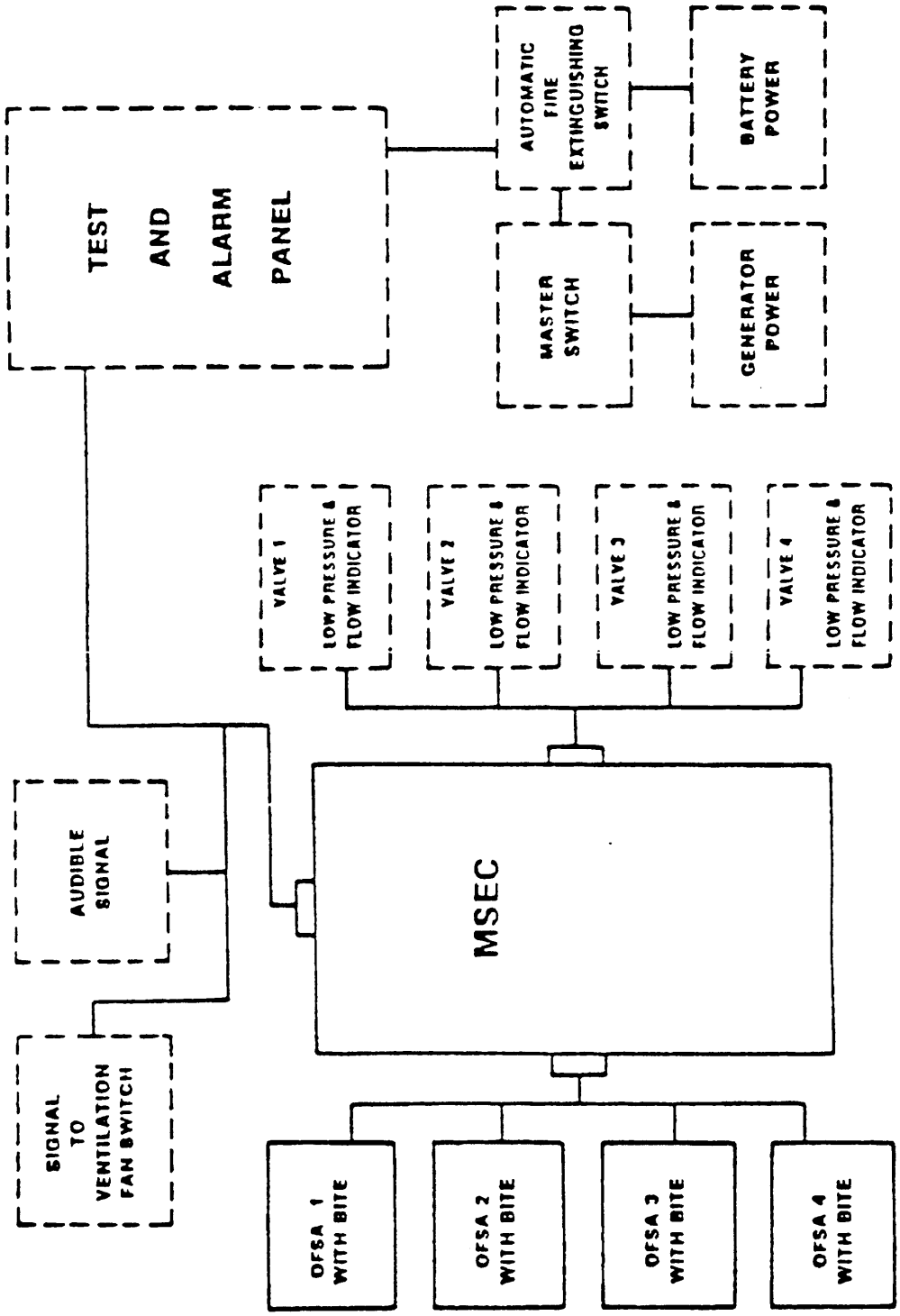


FIGURE 1. Block diagram example.

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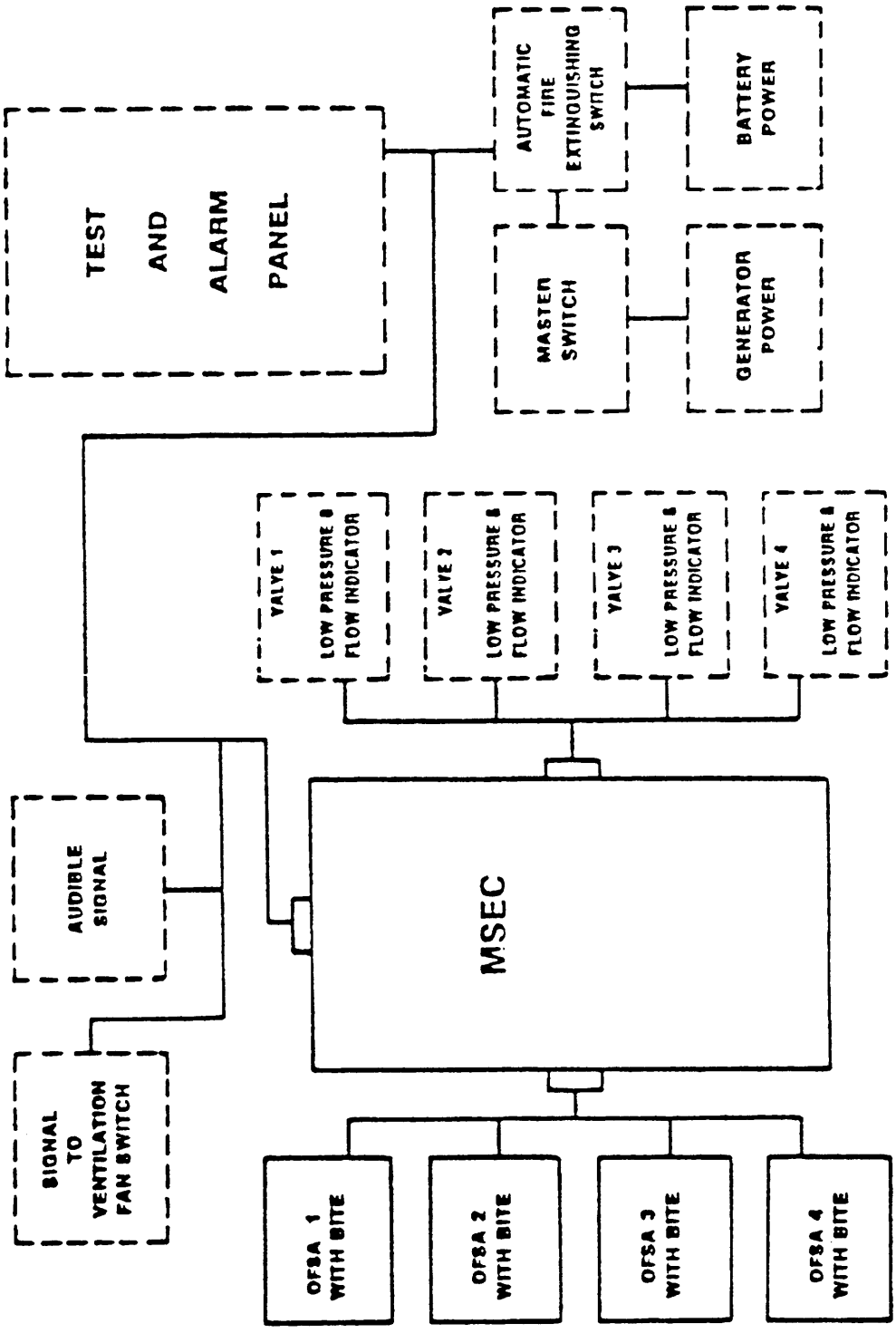


FIGURE 2. Block diagram example.

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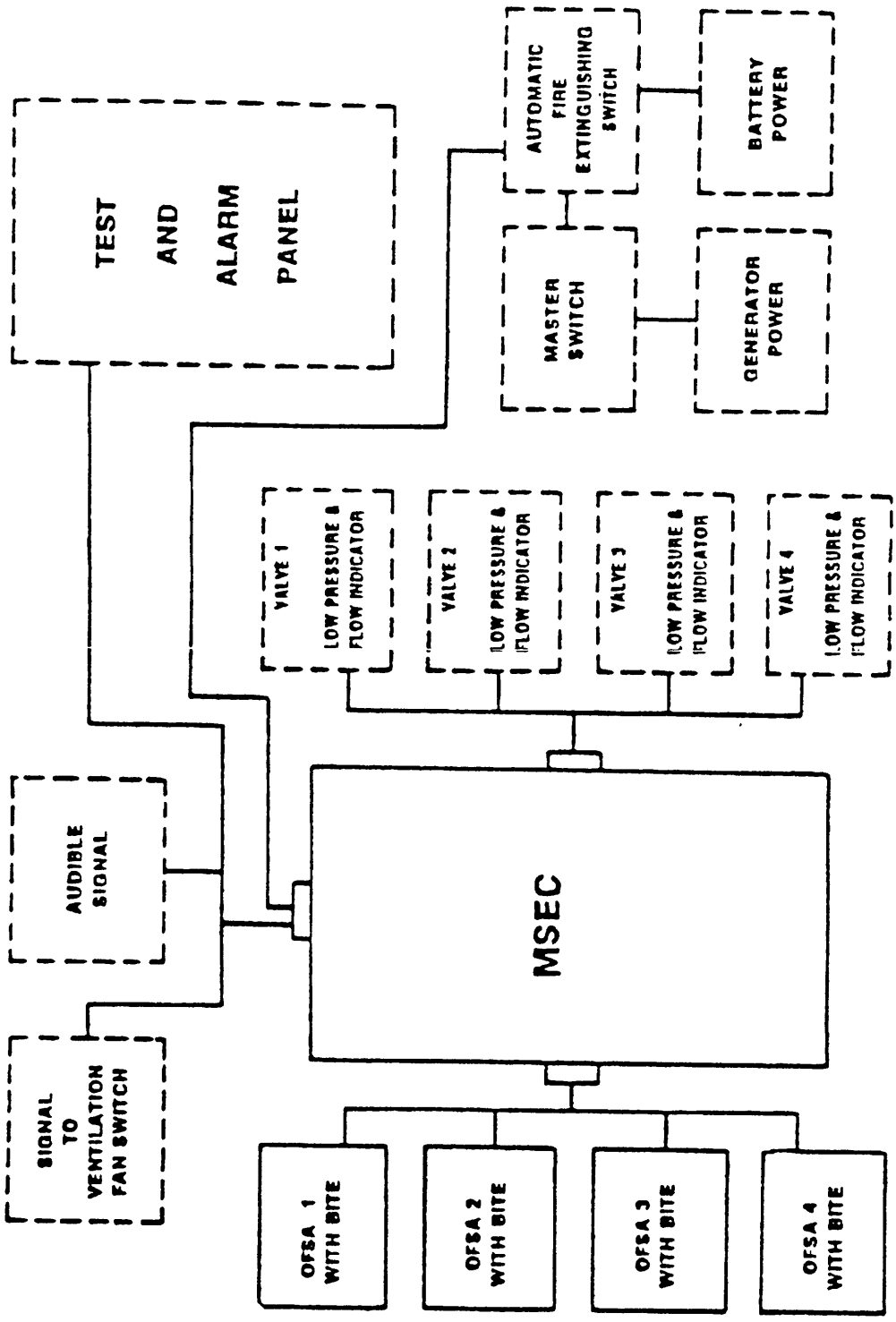
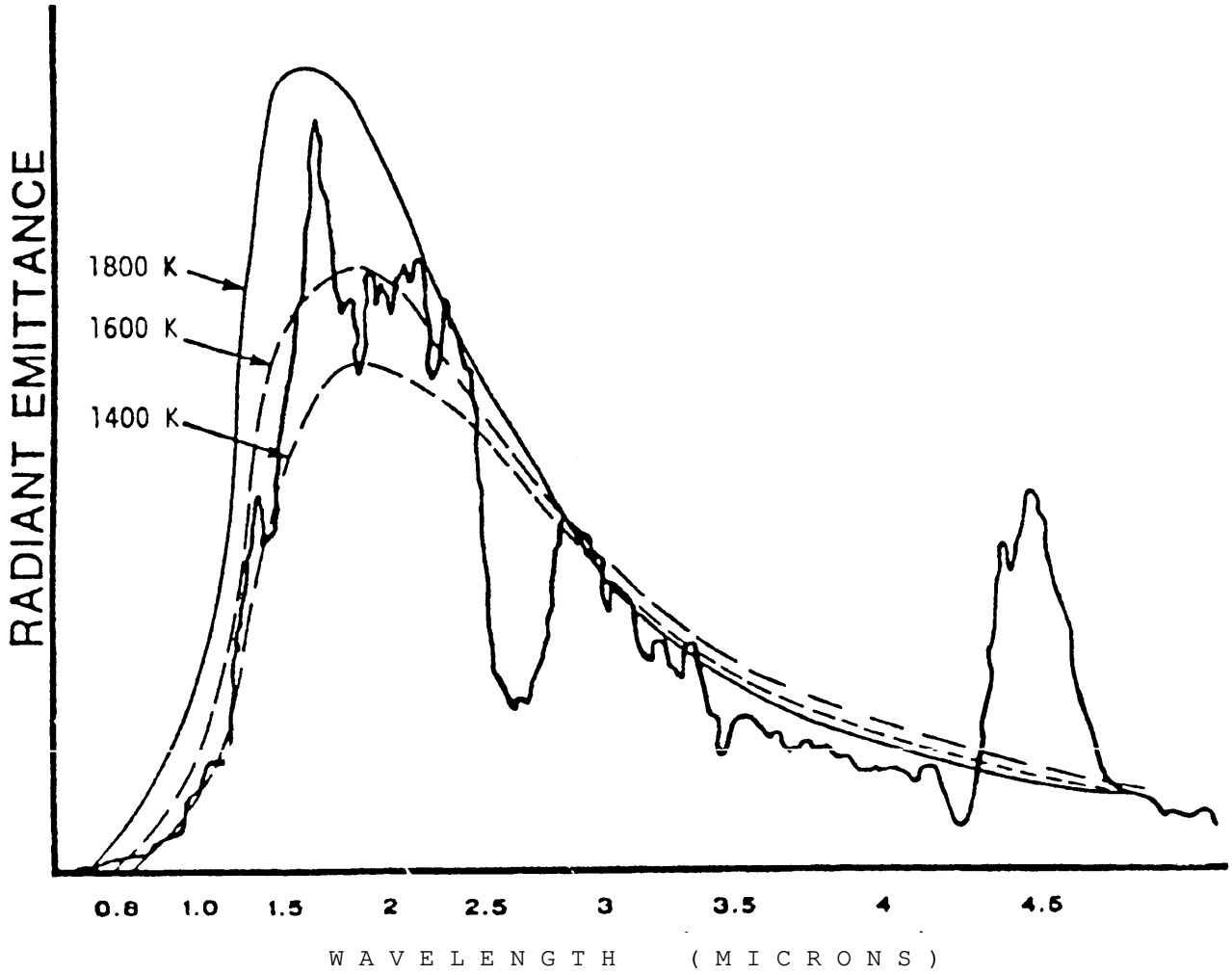


FIGURE 3. Block diagram example.

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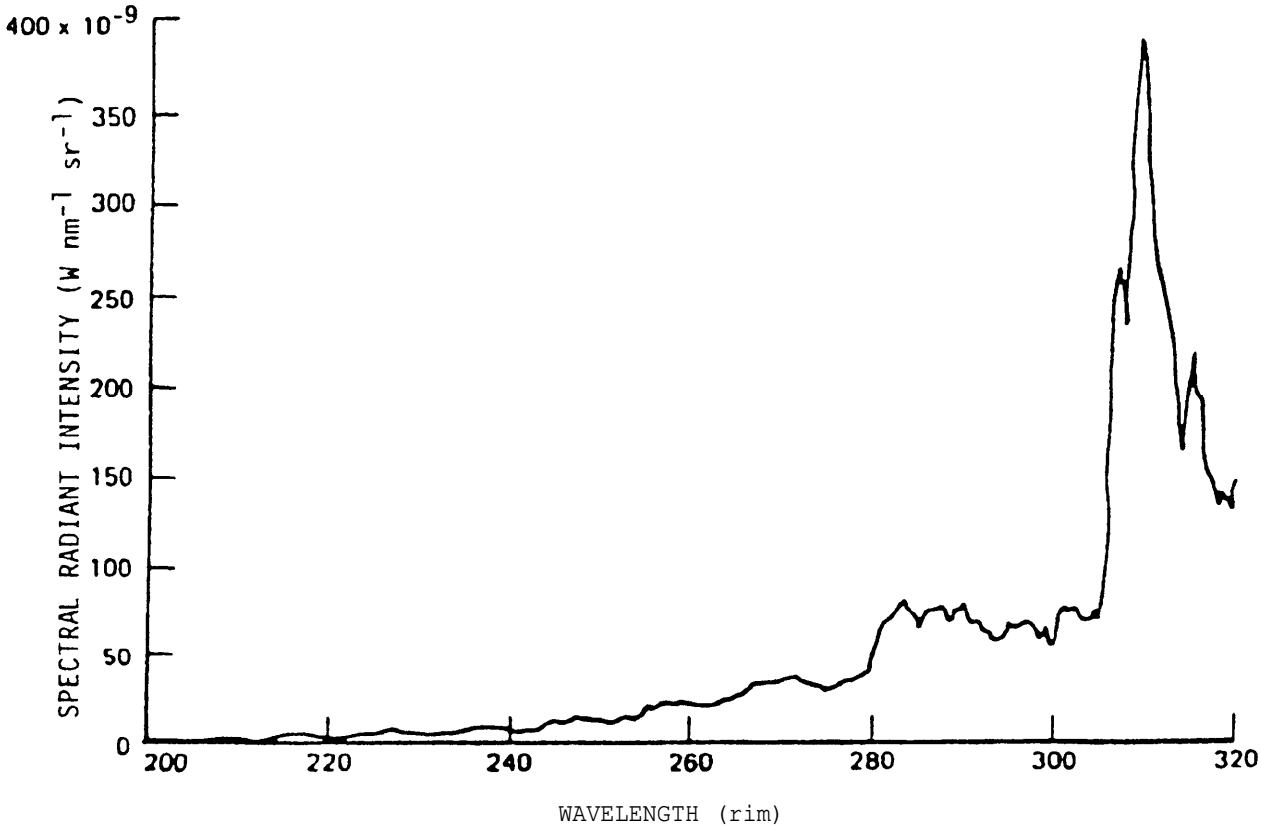
NOTES:

1. Kelvin (K).

2. Reference: Field Calometry/Chemical Studies by J. Kahrs and R.H. Tromans, December 1965, AD 368375.

FIGURE 4. Flame spectrum with graybody comparisons [280-mm (11-in) pan-fire using NTS].

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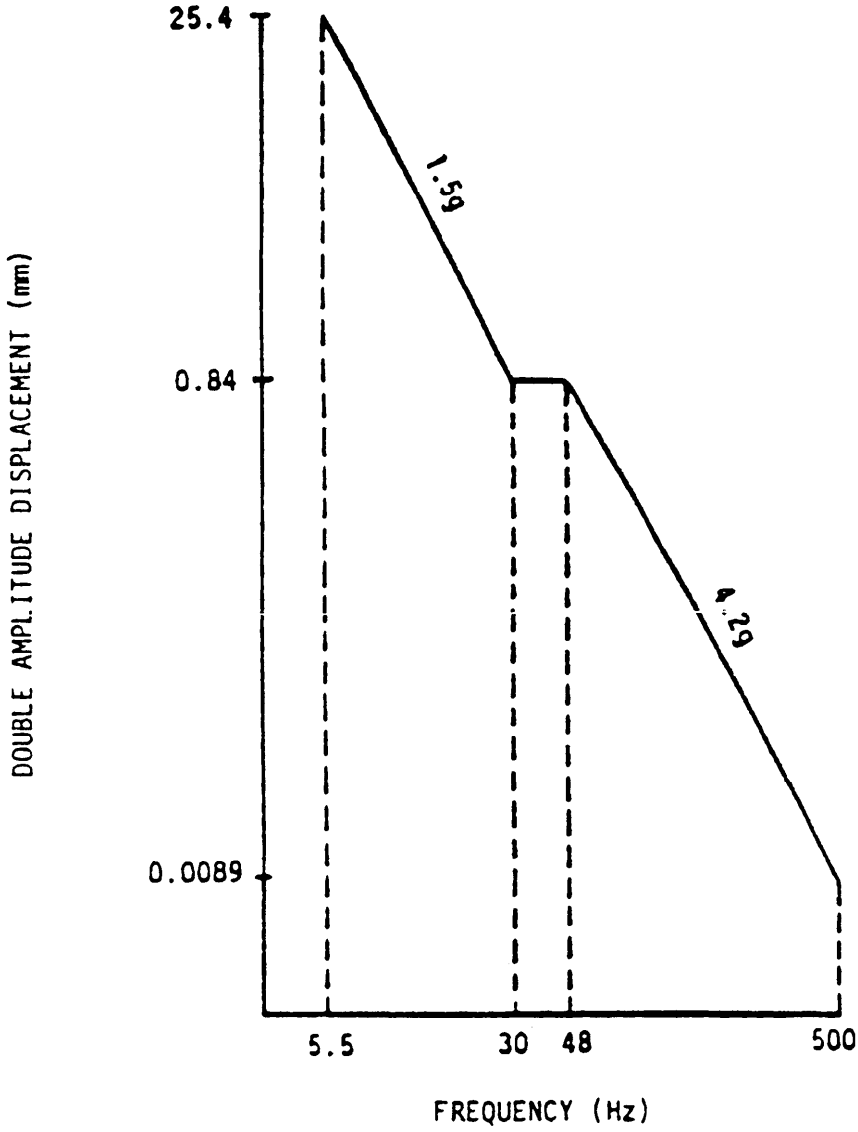


NOTES:

1. Nanometers (rim).
2. Watt per nanometer steradian (W nm^{-1} sr^{-1}).
3. Reference: Optical Emission Properties of Aircraft Combustible Fluids, August 1973, AFAPL-TR-73-83.

FIGURE 5. The ultraviolet emissions from JP-4 burning at sea level.

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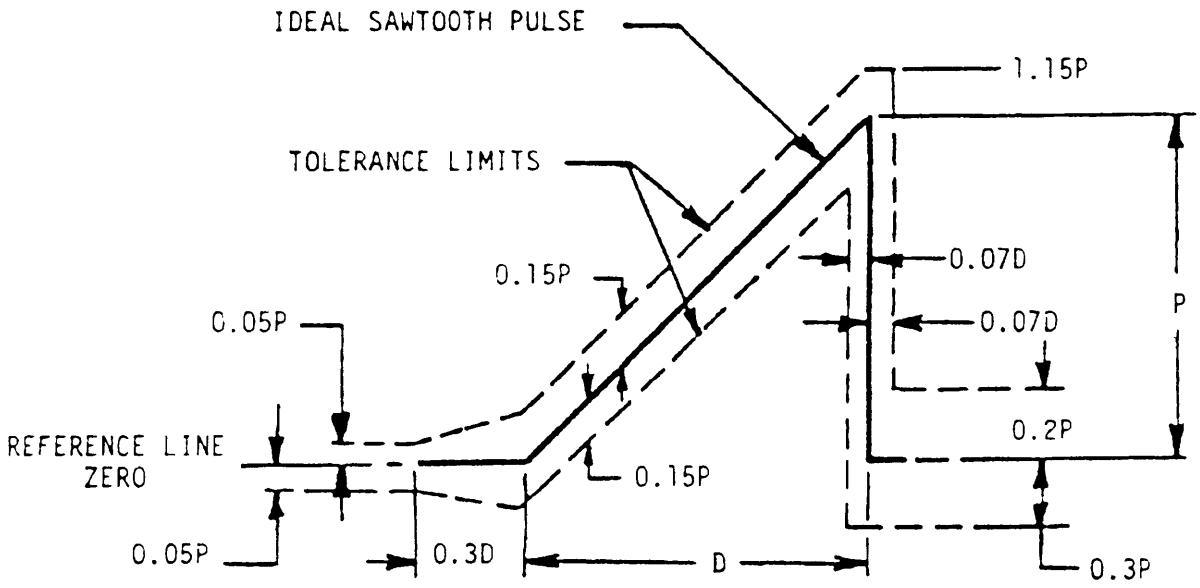


NOTES:

1. Hertz (Hz).
2. Gravity units (g).

FIGURE 6. Vibration test curve.

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NOTES:

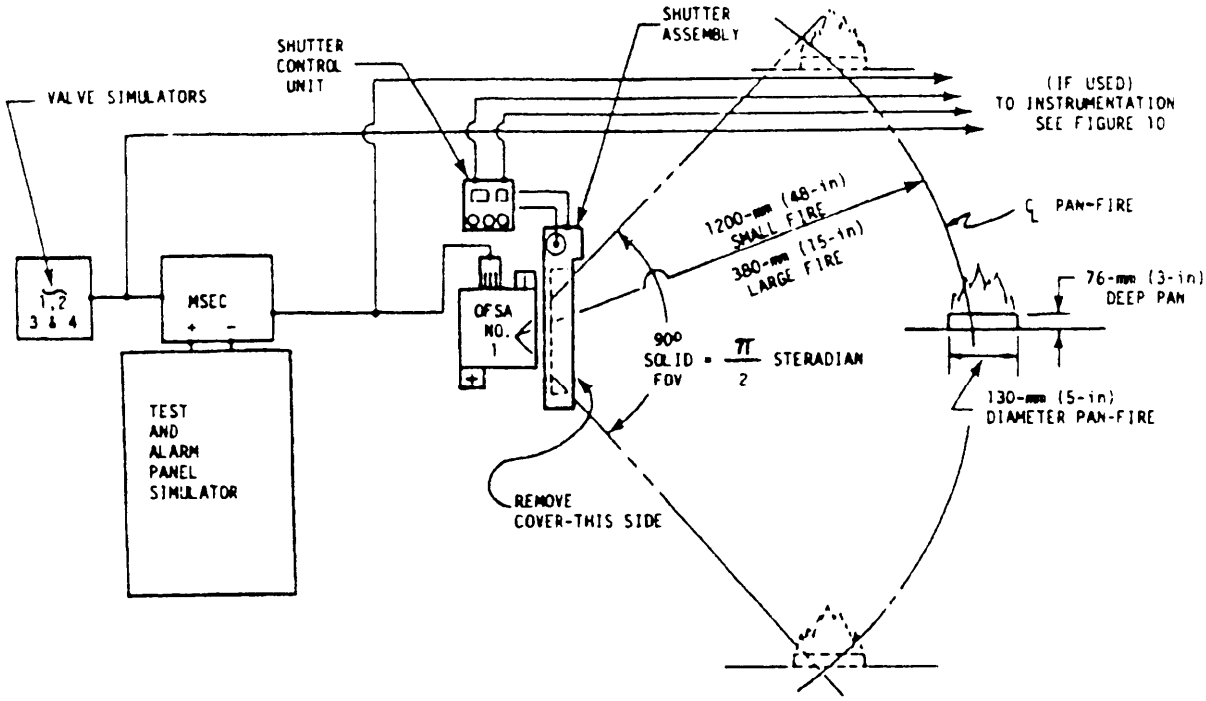
1. The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi $\pm$ 0.1 V_i, where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

2. Peak value (P) = 40g.

3. Nominal duration (D) = 11 ms.

FIGURE 7. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

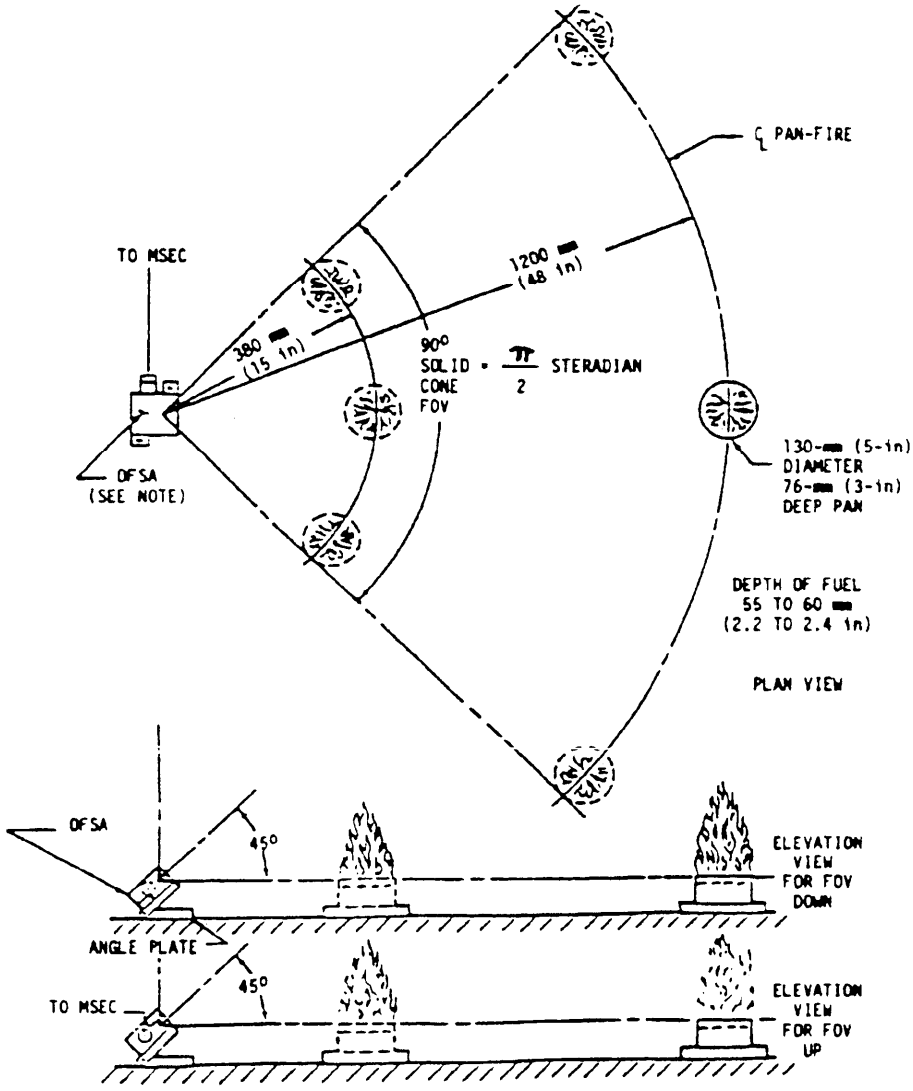
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NOTE : Use of shutter assembly and shutter control unit is optional (see figure 9).

FIGURE 8. Small fire (early fire warning) and large fire (secondary sensitivity threshold).

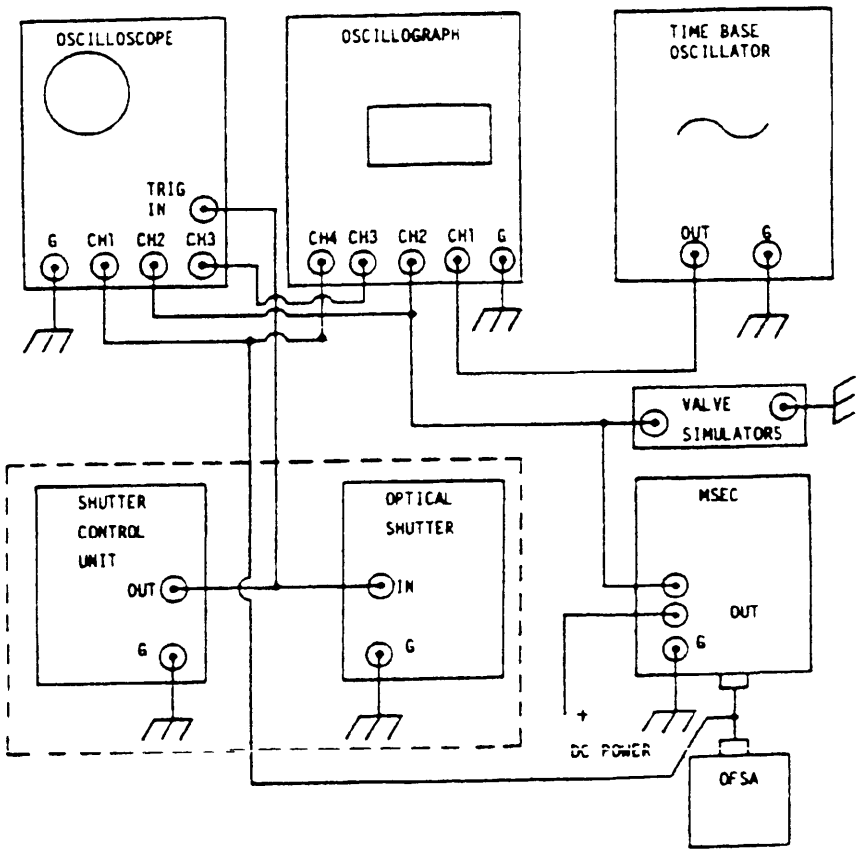
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NOTE : Optional OFSA may be rotated.

FIGURE 9. Small fire (early fire warning) and large fire (secondary sensitivity threshold).

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NOTES:

1. Use of shutter assembly and shutter control unit is optional.
2. Channel (CH).
3. Ground (G).

FIGURE 10. Electrical connections for small fire (early fire warning) and large fire (secondary sensitivity threshold).

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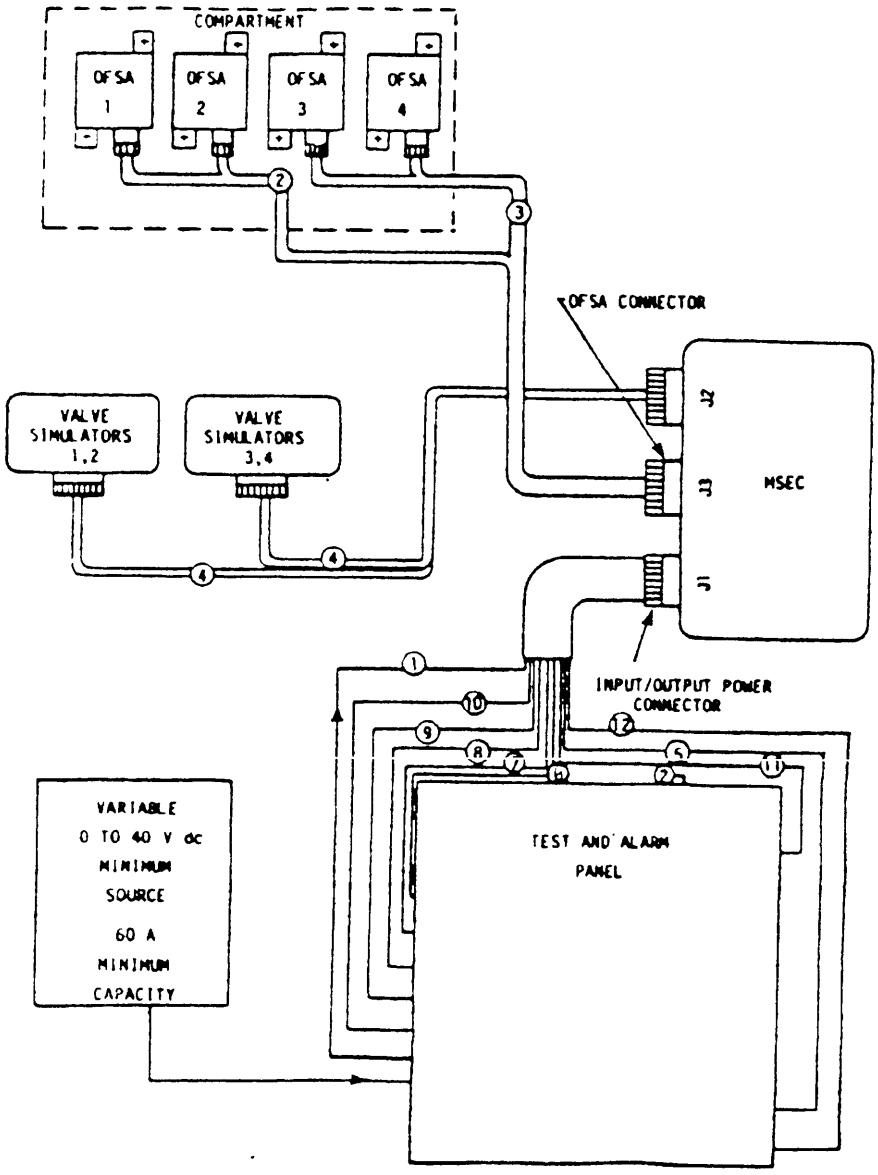


FIGURE 11. Input current tests.

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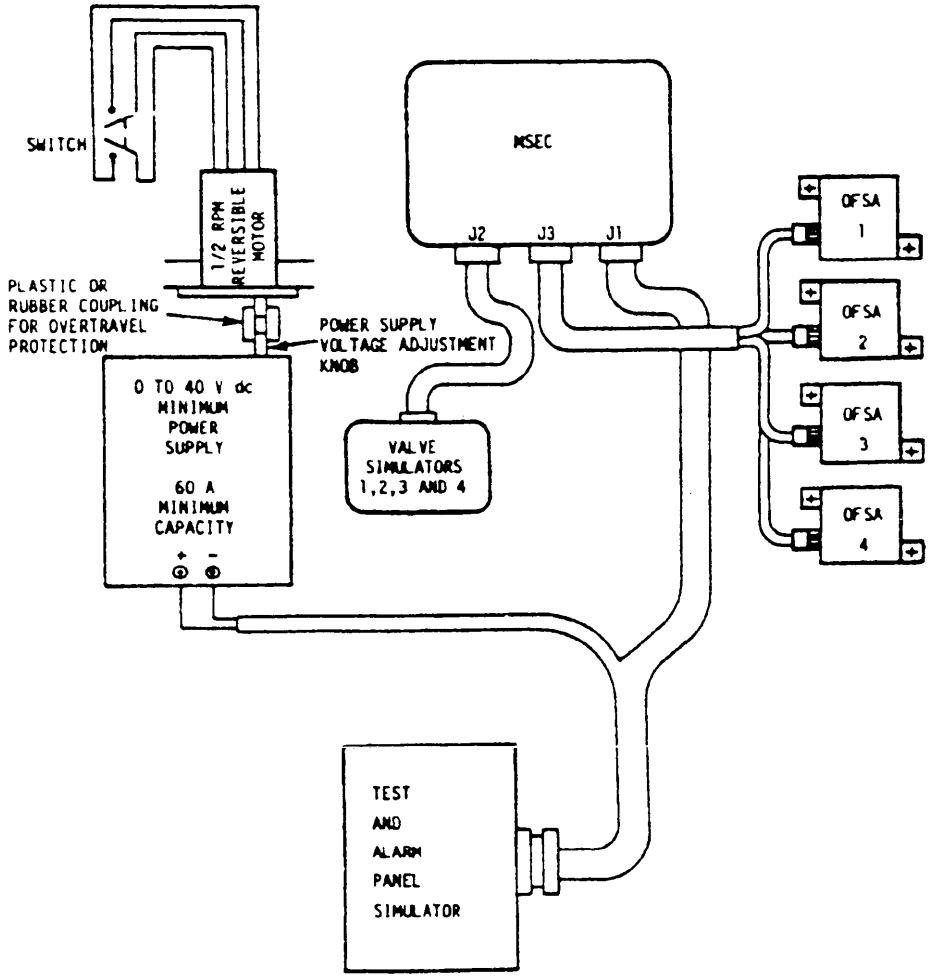


FIGURE 12. Voltage decay test.

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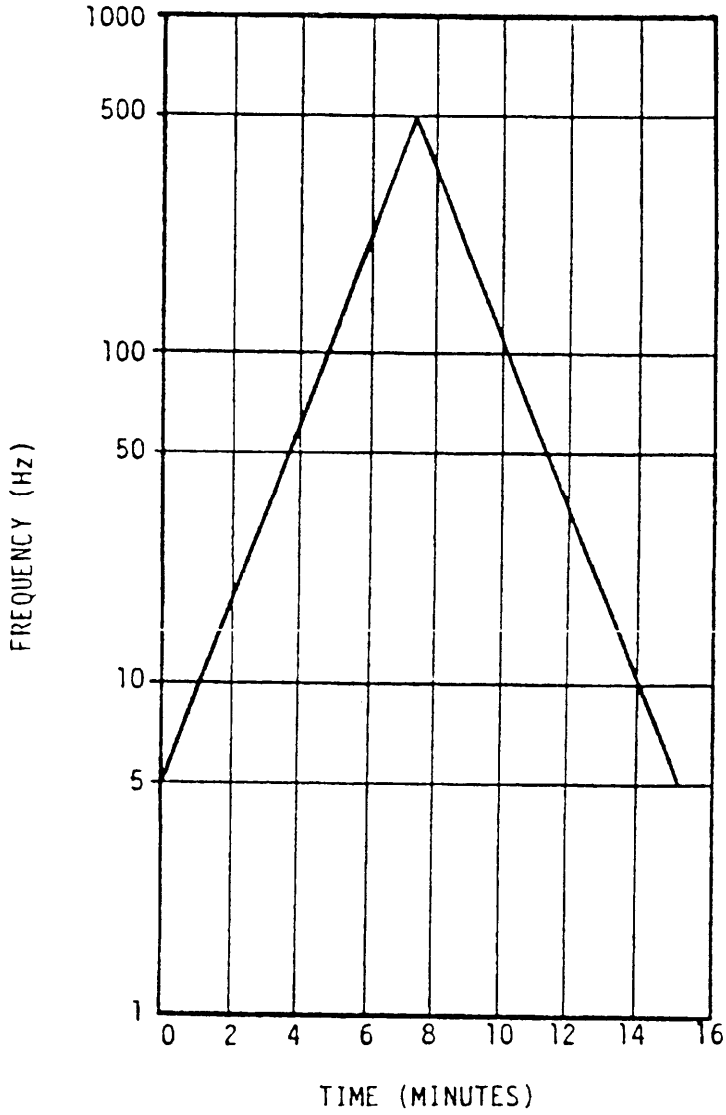
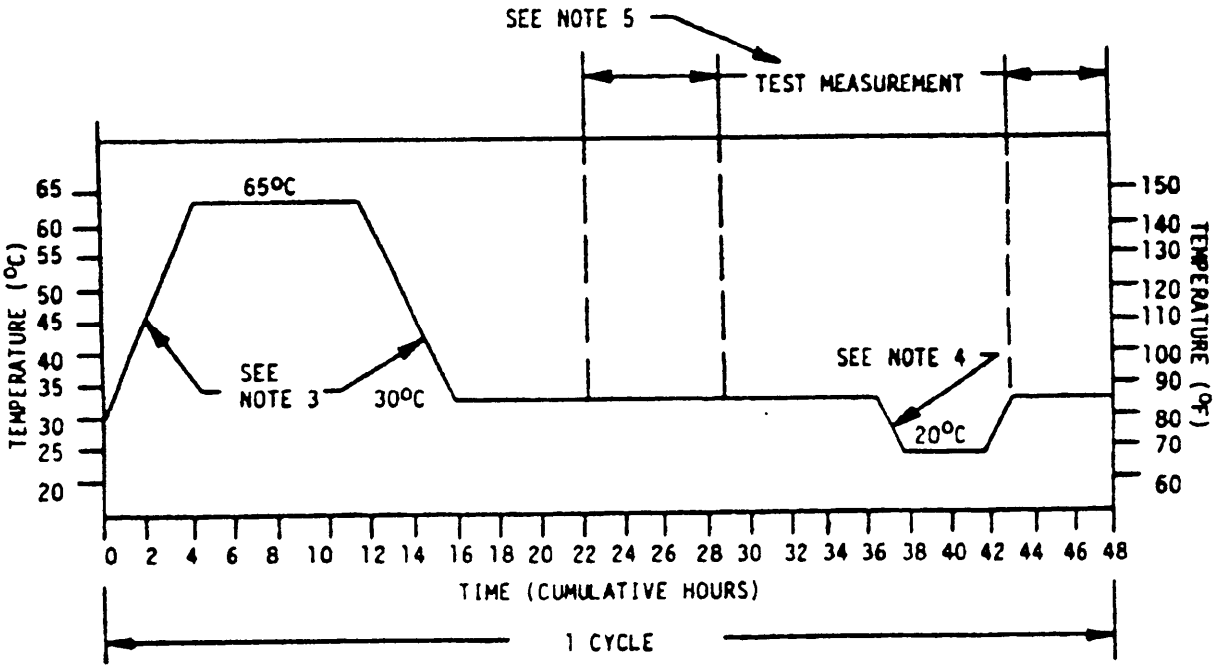


FIGURE 13. Logarithmic sweep.

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NOTES:

1. Tolerance during temperature change shall be not greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 30 and 65°C (86 and 149°F) shall be not less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion. of the curve shall be not less than 10°C (18°F).
5. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 14. Humidity cycle.

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